DEPARTMENT OF INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Robust Optimization of Charging Station Placement for Electric Trucks in Grocery Retail

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I confirm that this master's thesis in informatics is my own work and I have documented all sources and material used.

Munich, 04.05.2026 Max Elia Schweigkofler

Abstract

A retailer that wants to replace diesel trucks with battery-electric trucks faces a planning problem with long-term decisions and uncertain daily demand. Installing chargers at stores is expensive and hard to reverse once the chargers are in place. Demand, however, changes from day to day, so a plan that works for an average day may fail on a difficult day. For this reason, the thesis studies the infrastructure question as a worst-case planning problem: the chargers are placed so that even the hardest observed demand day can still be served, and so that the cost on that worst day is as low as possible. Minimizing the worst case instead of the average is the natural objective when the infrastructure decision is large, irreversible, and expected to hold on every day the retailer will face, not only on a typical one.

To make this concrete, historical delivery records from a large German grocery retailer are cleaned and transformed into optimization-ready instances with store coordinates, distance matrices, daily demand scenarios, and empirical unloading-time estimates. The charging station placement problem is then formulated as a two-stage robust mixed-integer program. In the first stage, the planner decides where to install store-site chargers; in the second stage, after a demand scenario is realized, the fleet is routed and charged based on that infrastructure, respecting payload, state of charge, and charging opportunities. Store-site chargers have an advantage in this setting: trucks already stop at each store to unload, so that idle time can be used for en-route charging. This opportunity charging extends the effective range of each truck and thereby enables longer tours. It also relaxes the dependence on warehouse charging, which is practically constrained: a large fleet charging simultaneously at the depot can quickly exceed what the local grid connection allows, creating a bottleneck that store-distributed charging avoids. Every parameter of the model is calibrated individually from technical truck specifications, historical delivery data, and cost assumptions for charging infrastructure, electricity, waiting, and truck use. The representative vehicle classes are the Mercedes-Benz eActros 300 and the Volvo FM Electric.

Since the resulting robust model is already computationally difficult for medium-sized instances, the thesis does not stop at the model formulation and also studies how the model can be solved in practice for larger instances. The main computational result is that direct full solves become impractical quite quickly, so clustering has to be treated as a core part of the algorithm. The implemented clustering family therefore includes a simple geographic

method, store-count angular slices, demand-aware angular slices, and a tour-containment variant that preserves whole historical tours as far as possible. The four methods go from simple to more operations-aware: geographic clustering only looks at location; store-count angular slices cut the map into equal-size pie slices around the depot; demand-aware angular slices cut the same slices but balance demand instead of store count; and tour-containment tries to keep whole historical tours inside the same cluster. A comparison on a 68-store test instance shows two clear groups: the two demand- or tour-aware methods give the lowest worst-case cost, while the two geometry-only methods are about 9 % more expensive on the worst scenario. Demand-aware angular slices are used as the default method, because they give the same cost as tour containment but are simpler to build, do not rely on historical tour data, and always give the same result for the same input, which ensures reproducibility, unlike the geographic k-means method whose outcome depends on the random initialization. Additional runtime experiments show that good first-stage solutions often stabilize much earlier than the reported MIP gap suggests, which supports tighter bounded-time budgets for easier clustered subproblems and longer budgets only for structurally harder ones. The final empirical study therefore uses the full available retailer instance with 259 stores, 30 historical demand scenarios, and demand-aware angular-slices clustering into 9 clusters. On this instance, the main comparison tests whether a moderate Stage 1 iteration budget is already enough or whether longer runs still change charger placement in a meaningful way.

The final outcome of the thesis is twofold. From a methodological view, it provides a calibrated robust Charging Station Placement Problem (CSPP) workflow that jointly optimizes charger placement and routing under a worst-case budget, and thereby decides whether installing store-side chargers is cheaper overall than relying on warehouse charging alone for electric grocery delivery. From a practical view, it integrates data preparation, clustering choice, runtime estimation, and the solve pipeline into a web-based decision-support tool that allows new store coordinates and demand scenarios to be imported and analyzed with the same planning logic. Broader reoptimization tests and cost-parameter sweeps are also carried out and reported as part of the thesis, showing how the placement decision reacts to changes in the underlying cost assumptions.

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1 Introduction

A retailer that wants to replace diesel trucks with battery-electric trucks faces a planning problem with long-term decisions and uncertain daily demand. If chargers are installed at stores, this is an infrastructure decision that is expensive and hard to reverse. Demand, however, changes from day to day. So a plan that works for an average day may fail on a difficult day. For this reason, the thesis studies a worst-case planning problem.

The thesis starts from real delivery data of a large German grocery retailer. The data was recorded while the fleet still used diesel trucks. Before building an optimization model, it was necessary to understand what the data actually shows. The data analysis gives the vehicle class, the unloading-time model, and the daily demand scenarios used later in the optimization. With these inputs in place, the main planning question can be addressed: where should the retailer install store chargers?

Grocery delivery is a good case for electrification. The trucks always drive from the same depot to a subset of the same retailer-owned stores, so any charger installed at these sites is used only by the retailer's own fleet. Trucks already stop at stores for unloading, and that idle time can be used for charging. Recharging during these mandatory stops extends the effective range of each truck and enables longer tours that would otherwise exceed the battery's single-charge reach. Another motivation is the grid constraint at the warehouse: for a large fleet, recharging all trucks at the depot in parallel can quickly exceed the available grid connection power, requiring expensive infrastructure upgrades. Distributing charging across store sites spreads the electrical load geographically and reduces the peak demand at the depot.

The thesis models this problem as a two-stage robust optimization problem. In the first stage, the retailer decides where to install chargers before the actual demand day is known. In the second stage, routing and charging decisions react to the realized demand scenario. This leads to the CSPP. The model is solved over historical demand days so that the charging plan remains feasible even on the hardest observed day.

The main difficulty is computational. With many stores, arcs, trucks, tours, and scenarios, the robust model becomes very large. A direct solve is not computationally feasible. For this reason, the thesis introduces clustering as a core part of the solution pipeline. The contribution is therefore not a new exact solver for the full robust problem. It is a practical

three-stage decomposition that makes a real 259-store robust CSPP solvable under bounded runtime and still keeps the first-stage charger decision interpretable. While the thesis builds and validates this workflow on a concrete case study, it goes one step further. As a final result, the workflow is packaged into a working web application. This means the practical outcome is a tool that any retailer can use to calculate the costs and benefits of electrifying their own fleet, without having to use the command line or write code.

1.1 Motivation: Electrifying Grocery Delivery

Food-related trucking in Germany is large and still almost entirely fossil-fuel based. In 2024, food and beverages accounted for 50.0 billion tonne-kilometres by domestic trucks [1], and in 2023 the transport sector still drew 97.8% of its energy from fuels [2]. Road-freight CO_2 emissions were still 14.6% above their 1995 level in 2023 [3]. Electric trucks can lower both emissions and operating-cost volatility: depot charging uses stable industrial electricity tariffs while diesel prices remain linked to volatile oil markets [4, 5, 6]. Several policy instruments also support the transition: KfW Zuschuss 441 subsidizes non-public company charging infrastructure with up to 900 EUR per charge point and up to 45,000 EUR per site [7]; electricity used in electric road vehicles can be monetized via the THG-Quote [8]; and zero-emission trucks remain exempt from the truck toll until 30 June 2031 [9].

1.2 Research Questions

The thesis is organized around three research questions.

1. How can historical retailer delivery data be transformed into a two-stage robust CSPP whose first-stage charger decision optimizes for the worst observed demand days?
2. Which parameters drive the worst-case planning result, and how can realistic values be derived from vehicle specifications, empirical data, and planning assumptions?
3. Since the robust CSPP is computationally difficult, which clustering and runtime choices make the problem practically solvable without losing the economic meaning of the first-stage charger decision?

1.3 Contributions

The thesis makes five main contributions.

- It builds a data pipeline that turns historical delivery records into optimization-ready instances with store coordinates, daily demand scenarios, unloading-time estimates, and route information.
- It formulates a two-stage robust CSPP for electric grocery delivery with load-dependent energy use, opportunity charging during unloading, and multi-tour operations.
- It calibrates vehicle, infrastructure, and operating-cost parameters from technical data, empirical records, and explicit planning assumptions.
- It studies clustering implementing a family of methods that goes from simple to more operations-aware: a simple geographic method that only looks at location, store-count angular slices that cut the map into equal-size pie slices around the depot, demand-aware angular slices that cut the same slices but balance demand instead of store count, and a tour-containment variant that tries to keep whole historical tours inside the same cluster.
- It implements and evaluates a three-stage decomposition with clustered robust solves, full scenario evaluation, and active-set reoptimization, and shows that this produces a complete robust plan for the real 259-store instance.
- It builds a working web application as a final result. This tool allows any retailer to upload their own delivery data, calculate the required infrastructure, and see the costs and gains of electrifying their fleet without needing programming skills.

1.4 Thesis Structure

Chapter 2 introduces the retailer data, derives the vehicle and unloading assumptions, and defines the energy and infrastructure setting. Chapter 3 places the thesis in the literature on electric vehicle routing, robust optimization, and decomposition. Chapter 4 gives the robust CSPP formulation and its model-defining preprocessing. Chapter 5 explains where the concrete parameter values come from. Chapter 6 describes the solution method, including the scenario-addition algorithm with z-bound and dominance pruning, the clustering pipeline, and the active-set reoptimization stage, traced along a concrete illustrative example cluster. Chapter 7 reports the empirical runtime calibration that fixes the bucket-based time limits used in all later experiments. Chapter 8 reports all experimental results: it compares four clustering methods on a 96-store instance, presents a cost-parameter sensitivity sweep on a single 16-store cluster, defines the full-instance experimental setup with the two cluster sizes that are run, briefly describes the web application add-on, and reports the completed

full-instance results (the exploratory run with 12 stores per cluster and the main 24-stores-per-cluster run). Chapter 9 discusses the findings and limitations. Chapter 11 closes the thesis.

2 Problem Setting and Data Basis

2.1 The Retailer Data

This chapter explains the retailer setting, the main observations from the data, and the assumptions derived from them. These assumptions form the basis of the optimization model in Chapter 4.

The starting point is one month of real delivery records from a large German grocery retailer in March 2025. At that time, the fleet still used diesel trucks. The raw records contain 28,924 entries. After cleaning, merging duplicates, normalizing time fields, and assigning coordinates, the processed dataset contains 6,775 tour stops, 259 stores, and 2,186 tours across 31 demand scenarios.

The first important observation is the load profile. Trucks leave the warehouse loaded, serve a sequence of stores, and then return almost empty. The median initial load per tour is about 9.5 tons, and the average tour serves about 4.2 stops. This fits medium-sized urban distribution trucks and not heavy long-haul trucks.

The second important observation is stop duration. At each store, the truck stands still during unloading. If the truck is electric, this waiting time can be used for charging. Because the stop happens regardless of whether a charger is present, en-route charging at stores adds energy without needing any additional stop for charging. This opportunity charging extends the effective daily range of each truck and allows tours that would otherwise exceed the battery's single-charge reach.

The third important observation is demand variation. Demand changes not only in total size but also in how it is distributed across stores. On some days demand is concentrated near the warehouse, while on other days demand is spread across a much broader area. This motivates the robust planning approach in Chapter 4.

2.2 Choosing Representative Electric Trucks

Based on the observed load profile, the thesis uses two representative electric trucks:

- Mercedes-Benz eActros 300 with payload capacity $L = 17{,}700$ kg and battery capacity $C = 336$ kWh,

- Volvo FM Electric with payload capacity $L = 16{,}200$ kg and battery capacity $C = 360$ kWh.

These vehicles represent two realistic truck classes with different payload and battery combinations. That is enough to make vehicle choice matter for both route feasibility and charger value. Both are marketed for urban distribution and grocery delivery [10, 11].

Other comparable electric trucks also exist, such as the Volvo FE Electric and FL Electric [12], the Renault Trucks E-Tech D and D Wide [13], and the MAN eTGL [14]. The method of the thesis does not depend on these exact product choices.

2.3 Unloading Time and Opportunity Charging

To model charging at stores, the thesis first estimates how long unloading takes. Using the cleaned stop history, unloading duration is estimated from 6,198 delivery stops as a linear function of delivered weight:

$$\text{Duration (min)} = 11.51 + 0.004184 \cdot \text{Weight (kg)}.$$

Stops with missing or implausible values were removed. The upper 1% of durations within each weight bin were also trimmed to remove extreme waiting-dominated outliers. Figure 2.1 shows the fitted regression.

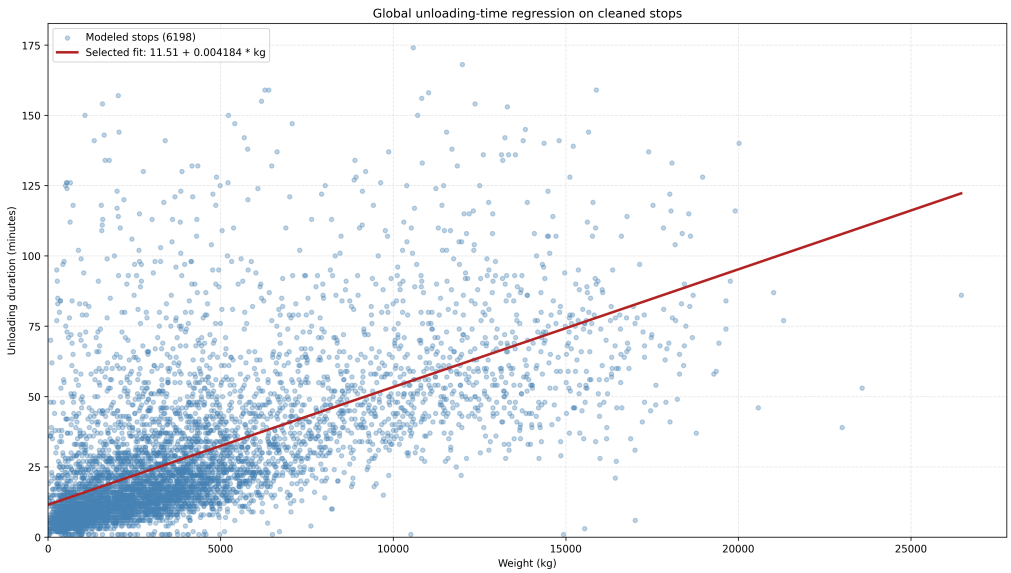

Figure 2.1: Unloading-time regression on 6,198 cleaned delivery stops. Each point is one stop; the red line is the fitted linear model.

The relationship is noisy, but the overall pattern is clear: heavier deliveries usually need

more unloading time. The energy charged during a stop is then modeled as charger power multiplied by the available time, with a linear efficiency factor [15]. This is a planning approximation. It does not try to model the full charging curve. It only translates stop duration into a possible charging amount.

2.4 Energy Consumption and State-of-Charge Assumptions

Energy consumption per kilometre is modeled as a linear function of the cargo load t on each arc:

$$P(t) = P_{\min} + \frac{t}{L}(P_{\max} - P_{\min}),$$

where $P_{\min}$ is consumption for an empty truck, $P_{\max}$ for a fully loaded truck, and L is payload capacity. Refrigeration and other secondary loads are not modeled explicitly. The numerical values of $P_{\min}$, $P_{\max}$, the State of Charge (SoC) window, and related parameters are specified in Chapter 5. The full battery propagation including store charging variables is given in Chapter 4.

2.5 Charging Infrastructure Assumptions

2.5.1 Charger Types and Installation Costs

The warehouse provides basic 22 kW AC charging in every configuration; this is kept fixed and is not a decision variable. This reflects a practical base setup: many trucks may need to charge in parallel at the depot, and the local grid may not support many high-power DC chargers at the same time. It is also mainly used for overnight charging, where slow charging is usually sufficient, and frequent fast charging can be more stressful for the battery [16]. Store chargers are the strategic decision of the model: the optimization decides which stores receive a charger and which charger type from the available set is installed there.

Store chargers have two main roles. First, they extend operating range by turning mandatory unloading stops into charging opportunities, enabling longer tours. Second, they reduce charging pressure at the warehouse by spreading the electrical load across the delivery network. If all charging is concentrated at the depot, the site can face peak loads that exceed the available grid connection, potentially requiring a costly upgrade to medium-voltage infrastructure [17, 18]; store-distributed charging avoids this bottleneck.

2.6 From Data to Optimization Instances

With these assumptions fixed, each historical day becomes one optimization instance. Each instance combines the active stores of that day, delivered weights, the distances between all possible locations (i.e. warehouse and stores), charger data, and vehicle parameters. The 31 historical days form the uncertainty set of the robust model, and the charger plan must remain feasible and be optimized for every one of them. Chapter 4 formalizes this optimization problem.

3 Related Work

This chapter discusses the related work that is most relevant for this thesis. The focus is on what the thesis takes from each paper and how the literature helps position the charger placement problem studied here.

3.1 Electric Vehicle Routing and Store Charging

Hiermann et al. [19] study a fleet-size-and-mix vehicle routing problem with time windows and charging stations for a heterogeneous electric fleet. They show that vehicle choice, battery limits, and routing decisions are tightly linked and should be optimized together instead of one after another. For this thesis, the key takeaway is that vehicle and routing decisions should be studied in one common framework.

Schiffer et al. [20] study electric commercial vehicle planning in a retail logistics setting. They show in a retail case study that infrastructure needs and operational performance can change strongly when battery assumptions, routing, and fleet composition are changed together. Their work therefore makes clear that isolated planning steps can give misleading results. For this thesis, the key takeaway is that grocery delivery needs an integrated planning view, even though the focus here is store-charger placement rather than fleet composition.

This thesis differs from standard electric routing papers in one important way. The main decision is not only how to route trucks, but where to install chargers at stores before the demand day is known. This matters because trucks already stop at stores for unloading, so they can charge there without an extra stop. Another practical issue is the depot grid limit. Borlaug et al. [17] show that charging many heavy-duty trucks at one depot can create a strong power peak in the local distribution grid. Their results show that depot charging can require substantial grid reinforcement once many trucks charge in parallel. For this thesis, the key takeaway is that store chargers are valuable for two reasons: they use unloading time for opportunity charging, and they reduce charging pressure at the warehouse.

3.2 Robust Planning and ASBP

The charger decision in this thesis is made before the actual demand day is known. Daily routing and charging decisions are made after the demand scenario is observed. This leads naturally to a two-stage robust model.

The direct methodological starting point of this thesis is recent work from the TUM Chair of Optimization and Sustainable Decision Making, where this thesis was written. Goerigk et al. [21] propose the approximate scenario bracketing procedure (ASBP) for finite-scenario two-stage robust mixed-integer problems. The method is built for settings where second-stage solves are expensive and only a small subset of scenarios is likely to be decisive. Its main idea is to work with scenario upper and lower bounds, screen weak scenarios early, and spend exact solve time only where it is still useful. This matches the CSPP in this thesis, because each second-stage problem is a mixed-integer routing-and-charging model and therefore costly to solve. ASBP is the direct algorithmic base of the solution method in this thesis.

Rahmaniani et al. [22] review decomposition methods for large two-stage optimization problems. Their review shows that such models are often too large for a direct solve and that practical performance depends on cuts, bounds, warm starts, and other acceleration ideas.

3.3 Clustering for Large Routing Problems

Cluster-first ideas have been used in routing for a long time. Fisher and Jaikumar [23] are a classic example of using decomposition to reduce routing complexity. This shows the general value of splitting a large routing problem into smaller parts. More recent work also shows that decomposition should preserve useful route structure, not only reduce problem size [24].

This is important here because clustering is the practical way to solve the robust CSPP at the retailer scale. The thesis therefore compares several clustering logics instead of assuming that one generic method is enough. The geographic method is the simplest baseline and mainly keeps nearby stores together. The store-count angular-slices method adds depot-centered radial structure. The demand-aware angular-slices method keeps the same radial idea but balances demand instead of store count. The tour-containment method tries to keep historical tours inside the same cluster. For this thesis, the key takeaway is that clustering is not only preprocessing; it is a central design choice that affects both runtime and solution quality.

3.4 Positioning of This Thesis

This thesis studies a retailer-specific charging station placement problem for electric grocery delivery under uncertain daily demand. The main decision is which stores should receive

chargers before the demand day is known. The model uses the fact that trucks already stop at stores for unloading, so these stops can also be used for charging. It also reflects that charging many trucks at the depot can be limited by the local grid.

Methodologically, the thesis combines a two-stage robust model with clustering and ASBP. ASBP is the robust solution method, and clustering is the decomposition step that makes the problem solvable at the retailer scale. In this sense, the thesis is not only about electric routing and not only about robust optimization. It studies how charger placement, uncertain demand, clustering, and bounded-time robust solving fit together in one practical grocery-delivery setting.

4 Charging Station Placement Problem

4.1 Problem Setting and Worst-Case Objective

The retailer chooses once where to install store chargers, and then operates deliveries every day under this fixed charger network. The open question for the model is which cost the charger decision should minimize. An average over all observed days would let a plan look good even if a few days are very expensive or even infeasible. On such heavy days the fleet may not finish its tours, and exactly these days drive the operational risk of the transition. The charger network should therefore be chosen so that the hardest observed day is still manageable.

Formally, let a denote a charger placement, $f(a)$ the one-time infrastructure cost, and $g^s(a)$ the operational cost on demand scenario $s \in S$. The worst-case daily operating cost is

$$z(a) \;:=\; \max_{s \in S} g^s(a),$$

and the planner solves

$$\min_{a} \; f(a) + z(a).$$

So the charger network is chosen so that the most expensive day becomes as cheap as possible, while investment cost is still taken into account. An empty placement is a valid outcome; it means that warehouse-only charging is already enough under the chosen assumptions.

This is a *robust* objective in the sense of [22, 25]: the decision is protected against the worst realization inside an uncertainty set, without requiring a probability distribution over scenarios. Here the uncertainty set is the finite set S of historical daily demand patterns of the retailer. Each scenario is a real day that actually occurred in the past, and the resulting outcome is: the charger plan is feasible, and its cost is bounded, for every day in the dataset.

This gives a natural two-stage structure: the charger placement a is fixed once, and routing and charging are then chosen per scenario in response to a. The scenario-level subproblems are themselves mixed-integer routing-and-charging problems with payload, battery, and charging constraints. The constraints $z \geq g^s(a)$ for all $s \in S$ link them to the placement a and push z up to the cost of the worst day.

The rest of the chapter describes this structure formally. Section 4.2 gives the mathematical

formulation. Section 4.3 explains the two preprocessing steps that are part of the model definition. Chapter 5 then explains where the concrete parameter values come from, and Chapter 6 describes the solution method.

4.2 Mathematical Formulation

4.2.1 Problem Structure and Notation Mapping

The `ChargingStationPlacementProblem` is modeled as a two-stage robust mixed-integer program with a finite uncertainty set. This class of problems, denoted as (2-RO) in [21], has the general form

$$\min_{x \in \mathcal{X}} \left(f(x) + \max_{s \in S} \min_{y^s \in \mathcal{Y}^s(x)} g^s(x, y^s) \right) \tag{4.1}$$

where x represents the first-stage decisions, y^s the second-stage decisions for realized scenario $s \in S$, and $\mathcal{X}, \mathcal{Y}^s(x)$ the feasible sets. For the `ChargingStationPlacementProblem`, this mapping is:

- **First-stage decision** x: The charger installation vector a, where a_j^τ determines if a charger of type τ is deployed at store j.
- **First-stage objective** $f(x)$: The infrastructure investment cost $f(a)$.
- **Second-stage decision** y^s: The operational vector for scenario s, including routing (r^s), payload flow (t^s), recharging (p^s, p^s_{wh}), and scheduling variables.
- **Second-stage objective** $g^s(x, y^s)$: The scenario-specific operational cost $g^s(a, y^s)$, which depends on the installed infrastructure a via recharging limits and waiting times.

So the robust objective minimizes investment cost plus worst-case operational cost. An additional structural constraint prevents more than one charger type from being installed at the same store; this single-type restriction is formalized in Section 4.2.7.

4.2.2 Sets and Parameters

Network structure

i_0	single warehouse (depot)
J	set of stores
V	set of nodes ($V = \{i_0\} \cup J$)
A	set of arcs (subset of $V \times V$ with known distances)
l_{v_1,v_2}	length of arc (v_1, v_2) for $(v_1, v_2) \in A$

Warehouse parameters

κ_{wh}	charging speed of one warehouse charger (22 kW)
α_{load}	fixed loading-time component at warehouse (in hours)
γ_{load}	additional loading time per kg loaded at warehouse (in hours/kg)

Fleet parameters

$K_{\max}$	upper bound on the number of trucks; derived from worst-case demand (see Section 4.3.2)
$M_{\max}$	maximum number of tours per truck; fixed by the experiment setting `max_tours_per_truck`
L	vehicle load capacity (tons)
C	vehicle battery capacity (kWh)
c_0	target SoC (in kWh); overnight charging restores this level
$P_{\min}$	kWh consumption per km for empty truck
$P_{\max}$	kWh consumption per km for fully loaded truck
F	fixed cost per used truck

Charging station parameters

T	set of charger types
e^τ	fixed cost for opening a charger of type τ at any store node
κ^τ	charging speed of charger type τ (in kW)
d	cost per kWh
h	cost per hour of extra waiting time (beyond unloading)
M	big-M constant ($M = C$, the battery capacity)

Demand parameters

S	finite set of scenarios
β_j^s	demand of store j in scenario s
δ_j^s	unloading time at store j in scenario s (in hours), proportional to β_j^s

4.2.3 First-Stage Decision Variables

Variable	Domain	Description
a_j^τ	$\{0, 1\}$	1 if charger type τ is installed at store j, 0 otherwise

Note: In the computational implementation used for the experiments, the warehouse charger is treated as a fixed 22 kW AC charging option. The strategic first-stage decision is therefore the deployment of store chargers.

4.2.4 Second-Stage Decision Variables

Second-stage decision variables for scenario $s \in S$:

Variable	Domain	Description
Truck/tour activation variables		
y_k^s	$\{0, 1\}$	1 if truck k is used, 0 otherwise

Variable	Domain	Description
$u^s_{k,m}$	$\{0,1\}$	1 if tour m of truck k is used, 0 otherwise
Flow variables (per truck $k \in [K_{\max}]$, tour $m \in [M_{\max}]$)		
$t^{s,k,m}_{v_1,v_2}$	$\mathbb{R}_{\geq 0}$	tons of cargo transported on arc (v_1, v_2) for $v_1, v_2 \in V$
$r^{s,k,m}_{v_1,v_2}$	$\{0,1\}$	1 if arc (v_1, v_2) for $v_1, v_2 \in V$ is used, 0 otherwise
SoC variables (per truck $k \in [K_{\max}]$, tour $m \in [M_{\max}]$)		
$c^{s,k,m,\text{arr}}_j$	$\mathbb{R}_{\geq 0}$	SoC upon arrival at store j (before charging, in kWh)
$p^{s,k,m}_j$	$\mathbb{R}_{\geq 0}$	energy charged at store j (in kWh)
$c^{s,k,m}_{\text{dep}}$	$\mathbb{R}_{\geq 0}$	SoC when departing warehouse on tour (k, m) (in kWh)
$c^{s,k,m}_{\text{ret}}$	$\mathbb{R}_{\geq 0}$	SoC when returning to warehouse after tour (k, m) (in kWh)
$p^{s,k,m}_{\text{wh}}$	$\mathbb{R}_{\geq 0}$	energy charged at warehouse before tour (k, m) (in kWh)
Overnight charging variable (per truck $k \in [K_{\max}]$)		
$p^{s,k}_{\text{overnight}}$	$\mathbb{R}_{\geq 0}$	energy charged overnight to restore c_0 (in kWh)
Waiting time variables (per truck $k \in [K_{\max}]$, tour $m \in [M_{\max}]$)		
$\omega^{s,k,m}_j$	$\mathbb{R}_{\geq 0}$	extra waiting time at store j beyond unloading (in hours)
$\omega^{s,k,m}_{\text{wh}}$	$\mathbb{R}_{\geq 0}$	extra waiting time at warehouse beyond loading (in hours)

The kWh consumption per km for cargo load $t_{v_1,v_2}^{s,k,m}$ is given as:

$$P(t_{v_1,v_2}^{s,k,m}) = P_{\min} + \frac{t_{v_1,v_2}^{s,k,m}}{L}(P_{\max} - P_{\min})$$

4.2.5 Objective Function

The value of the first-stage objective f is given by the cost of installing chargers at stores:

$$f(a) := \sum_{j \in J} \sum_{\tau \in T} e^{\tau} \cdot a_j^{\tau}.$$

The second-stage objective g^s for a scenario $s \in S$ is given as:

$$g^s := \sum_{k=1}^{K_{\max}} \sum_{m \in \{1,\dots,M_{\max}\}} \sum_{j \in J} d \cdot p_j^{s,k,m} \tag{4.2}$$

$$+ \sum_{k=1}^{K_{\max}} \sum_{m \in \{1,\dots,M_{\max}\}} \sum_{j \in J} h \cdot \omega_j^{s,k,m} \tag{4.3}$$

$$+ \sum_{k=1}^{K_{\max}} \sum_{m \in \{1,\dots,M_{\max}\}} d \cdot p_{\text{wh}}^{s,k,m} \tag{4.4}$$

$$+ \sum_{k=1}^{K_{\max}} \sum_{m \in \{1,\dots,M_{\max}\}} h \cdot \omega_{\text{wh}}^{s,k,m} \tag{4.5}$$

$$+ \sum_{k=1}^{K_{\max}} d \cdot p_{\text{overnight}}^{s,k} \tag{4.6}$$

$$+ \sum_{k=1}^{K_{\max}} F \cdot y_k^s. \tag{4.7}$$

The six terms of the second-stage objective represent different operating costs. Term (4.2) is the electricity cost of charging at stores. Term (4.3) penalizes waiting time at stores beyond unloading time. Term (4.4) is the electricity cost of warehouse charging between tours. Term (4.5) penalizes extra waiting at the warehouse. Term (4.6) captures overnight charging back to the target state of charge c_0. Term (4.7) is the fixed cost per used truck.

Together, these terms create the main trade-off of the model. Installing store chargers increases first-stage cost $f(a)$ but can reduce second-stage cost g^s by enabling en-route charging. Store chargers have a natural structural advantage here: trucks already stop at every store to unload, so the mandatory waiting time at each stop can be used for charging without any penalty. This allows trucks to recover energy along the route and extend their effective daily range, enabling longer tours that would otherwise require an extra return to the depot for charging. Relying only on warehouse charging avoids infrastructure investment but may require shorter tours, more depot returns, more trucks, and more waiting time. Furthermore, for a large fleet, recharging all trucks simultaneously at the depot can exceed the capacity of the available warehouse grid connection, as discussed in Section 5.3. Distributing charging across store

sites reduces the peak electrical load at the depot and thereby relaxes this grid bottleneck. The robust objective chooses the charger configuration with the lowest total cost.

4.2.6 Constraints

In the following, $[n]$ denotes $\{1, \ldots, n\}$ for any $n \in \mathbb{N}$.

The second-stage feasible set combines routing, payload flow, battery propagation, store charging, warehouse charging between tours, and reserve requirements. So the formulation is easiest to read in blocks. Table 4.3 lists what each block enforces before the full constraints appear.

Table 4.3: Second-stage constraint blocks and what they enforce.

Block	What it enforces
Symmetry breaking	Trucks are used in index order to avoid equivalent permutations.
Truck and tour activation	A tour can only be used if its truck is active; tours of one truck are used in order without gaps.
Tour structure	Each active tour starts and ends at the warehouse.
Flow constraints	Every store with positive demand is visited exactly once; cargo flow is conserved and bounded by payload capacity.
Energy constraints	The battery state of charge is propagated along each arc; consumption depends on the cargo load $P(t)$.
Overnight charging	After the last tour, each truck is charged back to the target SoC c_0.
Charging constraints	Charging at a store is only possible if a charger is installed there; warehouse charging happens between tours.
SoC bounds	The battery stays between 20% and 100% of capacity.

Given a scenario $s \in S$ and a first-stage solution a, the feasible set $\mathcal{Y}^s(a)$ is defined by the following constraints:
The symmetry-breaking constraint orders trucks by index. Truck $k+1$ can only be active if truck k is already active. Without this rule, the solver would explore many equivalent truck permutations.

Symmetry breaking constraint:

$$y^s_{k+1} \leq y^s_k \qquad \forall k \in [K_{\max} - 1] \tag{4.8}$$

The truck and tour activation constraints link tours to trucks. Constraint ($\forall k \in [K_{\max}], m \in [M_{\max}]$ (4.9)) says that a tour can only be used if its truck is active. Constraint ($\forall k \in [K_{\max}], m \in [M_{\max} - 1]$ (4.10)) says that tours of one truck are used in order, so tour $m+1$ is only used if tour m is also used. Together they remove gaps in the tour indices.

Truck and tour activation constraints:

$$u^s_{k,m} \leq y^s_k \qquad \forall k \in [K_{\max}], m \in [M_{\max}] \quad (4.9)$$

$$u^s_{k,m+1} \leq u^s_{k,m} \qquad \forall k \in [K_{\max}], m \in [M_{\max} - 1] \quad (4.10)$$

The tour structure constraints require each active tour to start and end at the warehouse. Constraint ($\forall k \in [K_{\max}], m \in [M_{\max}]$ (4.11)) forces exactly one arc from the warehouse to some store at the start of an active tour, and constraint ($\forall k \in [K_{\max}], m \in [M_{\max}]$ (4.12)) forces exactly one arc from some store back to the warehouse at the end. For inactive tours, both sums are zero. This reflects the depot-return structure of grocery delivery.

Tour structure constraints:

$$\sum_{j \in J} r^{s,k,m}_{i_0,j} = u^s_{k,m} \qquad \forall k \in [K_{\max}], m \in [M_{\max}] \quad (4.11)$$

$$\sum_{j \in J} r^{s,k,m}_{j,i_0} = u^s_{k,m} \qquad \forall k \in [K_{\max}], m \in [M_{\max}] \quad (4.12)$$

The flow constraints cover both the visits and the cargo on board. Constraint ($\forall j \in J : \beta^s_j > 0$ (4.13)) forces each store with positive demand to be entered exactly once across all trucks and tours. Constraint ($\forall j \in J : \beta^s_j = 0$ (4.14)) forbids visits to stores with zero demand. Constraint ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.15)) is the routing flow balance: if a truck enters a store on a tour, it must also leave it on the same tour. Constraint ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.16)) is the payload flow balance: the cargo arriving at a store equals the cargo leaving it plus the demand delivered there. Constraint ($\forall v_1, v_2 \in V, k \in [K_{\max}], m \in [M_{\max}]$ (4.17)) limits the cargo on any arc to the truck capacity L and also forces cargo to zero on arcs that are not used. The cargo-flow variable t therefore links routing with energy consumption via $P(t)$.

Flow constraints:

$$\sum_{k=1}^{K_{\max}} \sum_{m=1}^{M_{\max}} \sum_{v \in V \setminus \{j\}} r_{v,j}^{s,k,m} = 1 \qquad \forall j \in J : \beta_j^s > 0 \quad (4.13)$$

$$\sum_{k=1}^{K_{\max}} \sum_{m=1}^{M_{\max}} \sum_{v \in V \setminus \{j\}} r_{v,j}^{s,k,m} = 0 \qquad \forall j \in J : \beta_j^s = 0 \quad (4.14)$$

$$\sum_{v \in V \setminus \{j\}} r_{v,j}^{s,k,m} = \sum_{v \in V \setminus \{j\}} r_{j,v}^{s,k,m} \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.15)$$

$$\sum_{v \in V \setminus \{j\}} t_{v,j}^{s,k,m} = \sum_{v \in V \setminus \{j\}} t_{j,v}^{s,k,m} + \beta_j^s \cdot \sum_{v \in V \setminus \{j\}} r_{v,j}^{s,k,m} \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.16)$$

$$t_{v_1,v_2}^{s,k,m} \leq L \cdot r_{v_1,v_2}^{s,k,m} \qquad \forall v_1, v_2 \in V, k \in [K_{\max}], m \in [M_{\max}] \quad (4.17)$$

The energy constraints propagate the battery state of charge along each tour. Constraint ($\forall k \in [K_{\max}]$ (4.18)) sets the departure SoC of the first tour to the target level c_0, and to zero if the truck is unused. The pairs ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.19))–($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.20)), ($\forall v_1, v_2 \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.21))–($\forall v_1, v_2 \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.22)) and ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.23))–($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.24)) each enforce an equality in big-M form only on the arcs that are actually used: ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.19))–($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.20)) links the departure SoC at the warehouse with the arrival SoC at the first store, ($\forall v_1, v_2 \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.21))–($\forall v_1, v_2 \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.22)) links consecutive store visits and also adds the energy charged at the previous store, and ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.23))–($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.24)) links the last store with the return SoC at the warehouse. The pair ($\forall k \in [K_{\max}], m \in [M_{\max} - 1]$ (4.25))–($\forall k \in [K_{\max}], m \in [M_{\max} - 1]$ (4.26)) connects one tour to the next: the departure SoC of tour $m+1$ equals the return SoC of tour m plus the energy charged at the warehouse in between. The key point is that energy consumption depends on the cargo load via $P(t)$, so a loaded truck consumes more energy per kilometre than an empty one.

Energy constraints:

$$c_{\text{dep}}^{s,k,1} = c_0 \cdot u_{k,1}^s \qquad \forall k \in [K_{\max}] \quad (4.18)$$

$$c_j^{s,k,m,\text{arr}} \leq c_{\text{dep}}^{s,k,m} - l_{i_0,j} \cdot P(t_{i_0,j}^{s,k,m})$$

$$+ M(1 - r^{s,k,m}_{i_0,j}) \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.19)$$

$$c^{s,k,m,\text{arr}}_j \geq c^{s,k,m}_{\text{dep}} - l_{i_0,j} \cdot P(t^{s,k,m}_{i_0,j})$$
$$- M(1 - r^{s,k,m}_{i_0,j}) \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.20)$$

$$c^{s,k,m,\text{arr}}_{v_2} \leq c^{s,k,m,\text{arr}}_{v_1} + p^{s,k,m}_{v_1} - l_{v_1,v_2} \cdot P(t^{s,k,m}_{v_1,v_2})$$
$$+ M(1 - r^{s,k,m}_{v_1,v_2}) \qquad \forall v_1, v_2 \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.21)$$

$$c^{s,k,m,\text{arr}}_{v_2} \geq c^{s,k,m,\text{arr}}_{v_1} + p^{s,k,m}_{v_1} - l_{v_1,v_2} \cdot P(t^{s,k,m}_{v_1,v_2})$$
$$- M(1 - r^{s,k,m}_{v_1,v_2}) \qquad \forall v_1, v_2 \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.22)$$

$$c^{s,k,m}_{\text{ret}} \leq c^{s,k,m,\text{arr}}_j + p^{s,k,m}_j - l_{j,i_0} \cdot P(t^{s,k,m}_{j,i_0})$$
$$+ M(1 - r^{s,k,m}_{j,i_0}) \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.23)$$

$$c^{s,k,m}_{\text{ret}} \geq c^{s,k,m,\text{arr}}_j + p^{s,k,m}_j - l_{j,i_0} \cdot P(t^{s,k,m}_{j,i_0})$$
$$- M(1 - r^{s,k,m}_{j,i_0}) \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.24)$$

$$c^{s,k,m+1}_{\text{dep}} \leq c^{s,k,m}_{\text{ret}} + p^{s,k,m+1}_{\text{wh}} + M(1 - u^s_{k,m+1}) \qquad \forall k \in [K_{\max}], m \in [M_{\max} - 1] \quad (4.25)$$

$$c^{s,k,m+1}_{\text{dep}} \geq c^{s,k,m}_{\text{ret}} + p^{s,k,m+1}_{\text{wh}} - M(1 - u^s_{k,m+1}) \qquad \forall k \in [K_{\max}], m \in [M_{\max} - 1] \quad (4.26)$$

The overnight charging constraints refill the battery at the end of the day. Constraint ($\forall k \in [K_{\max}], m \in [M_{\max} - 1]$ (4.27)) activates for every truck after the tour m that is its last used tour (so $u^s_{k,m} - u^s_{k,m+1} = 1$), and forces the overnight energy to be at least the gap between c_0 and the return SoC of that tour. Constraint ($\forall k \in [K_{\max}]$ (4.28)) handles the case where tour $M_{\max}$ is itself the last tour. Constraint ($\forall k \in [K_{\max}]$ (4.29)) forces the overnight energy to zero for trucks that are not used. Together they ensure that every operating day starts from the same battery level c_0.

Overnight charging constraints:

$$p^{s,k}_{\text{overnight}} \geq c_0 \cdot (u^s_{k,m} - u^s_{k,m+1}) - c^{s,k,m}_{\text{ret}} \qquad \forall k \in [K_{\max}], m \in [M_{\max} - 1] \quad (4.27)$$

$$p^{s,k}_{\text{overnight}} \geq c_0 \cdot u^s_{k,M_{\max}} - c^{s,k,M_{\max}}_{\text{ret}} \qquad \forall k \in [K_{\max}] \quad (4.28)$$

$$p^{s,k}_{\text{overnight}} \leq C \cdot y^s_k \qquad \forall k \in [K_{\max}] \quad (4.29)$$

The charging constraints connect the first-stage charger decision with the second-stage operations. Constraint ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.30)) limits the energy charged at a store by charger power times available time (unloading plus extra waiting), and also forces it to zero if no charger is installed there. Constraints ($\forall k \in [K_{\max}]$ (4.31)) and ($\forall k \in [K_{\max}]$ (4.32)) set warehouse charging and warehouse waiting to zero before the first tour, because that tour already starts at the target SoC c_0. Constraint ($\forall k \in [K_{\max}], m \in \{2, \dots, M_{\max}\}$ (4.33)) limits the warehouse charging energy before tour $m \geq 2$ by charger power times the available window (fixed loading component, load-dependent loading component, and extra waiting). Constraint ($\forall k \in [K_{\max}], m \in \{2, \dots, M_{\max}\}$ (4.34)) forces warehouse waiting to zero for tours that are not used.

Charging constraints:

$$p_j^{s,k,m} \leq \sum_{\tau \in T} \kappa^\tau \cdot (\delta_j^s + \omega_j^{s,k,m}) \cdot a_j^\tau \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.30)$$

$$p_{\text{wh}}^{s,k,1} = 0 \qquad \forall k \in [K_{\max}] \quad (4.31)$$

$$\omega_{\text{wh}}^{s,k,1} = 0 \qquad \forall k \in [K_{\max}] \quad (4.32)$$

$$p_{\text{wh}}^{s,k,m} \leq \kappa_{\text{wh}} \cdot \left(\alpha_{\text{load}} \cdot u_{k,m}^s + \gamma_{\text{load}} \cdot \sum_{j \in J} t_{i_0,j}^{s,k,m} + \omega_{\text{wh}}^{s,k,m} \right) \qquad \forall k \in [K_{\max}], m \in \{2, \dots, M_{\max}\} \quad (4.33)$$

$$\omega_{\text{wh}}^{s,k,m} \leq M \cdot u_{k,m}^s \qquad \forall k \in [K_{\max}], m \in \{2, \dots, M_{\max}\} \quad (4.34)$$

The state-of-charge bounds keep the battery level between 20% and 100% of capacity. Constraint ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.35)) enforces the 20% floor on arrival at a store that is actually visited. Constraint ($\forall k \in [K_{\max}], m \in [M_{\max}]$ (4.36)) enforces the same floor for the return SoC of each used tour. Constraint ($\forall j \in J, k \in [K_{\max}], m \in [M_{\max}]$ (4.37)) forces the SoC after store charging to stay below full capacity C. Constraint ($\forall k \in [K_{\max}], m \in [M_{\max}]$ (4.38)) bounds the departure SoC by C for used tours and forces it to zero otherwise. The 20% floor avoids deep discharge.

SoC bounds:

$$c_j^{s,k,m,\text{arr}} \geq 0.2 \cdot C \cdot \sum_{v \in V \setminus \{j\}} r_{v,j}^{s,k,m} \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \quad (4.35)$$

$$c_{\text{ret}}^{s,k,m} \geq 0.2 \cdot C \cdot u_{k,m}^{s} \qquad \forall k \in [K_{\max}], m \in [M_{\max}] \tag{4.36}$$

$$c_{j}^{s,k,m,\text{arr}} + p_{j}^{s,k,m} \leq C \qquad \forall j \in J, k \in [K_{\max}], m \in [M_{\max}] \tag{4.37}$$

$$c_{\text{dep}}^{s,k,m} \leq C \cdot u_{k,m}^{s} \qquad \forall k \in [K_{\max}], m \in [M_{\max}] \tag{4.38}$$

4.2.7 The Full CSPP

The full model, reformulated as a minimization problem, then reads:

$$\begin{aligned}
\min \quad & f(a) + z \\
\text{s.t.} \quad & \sum_{\tau \in T} a_j^{\tau} \leq 1 && \forall j \in J \\
& z \geq g^s && \forall s \in S \\
& (\forall k \in [K_{\max} - 1] \quad (4.8)) - (\forall k \in [K_{\max}], m \in [M_{\max}] \quad (4.38)) && \forall s \in S \\
& a \in \{0,1\}^{J \times T},\ z \in \mathbb{R}.
\end{aligned} \tag{CSPP}$$

The variable z is the worst-case daily operating cost from Section 4.1. The constraints $z \geq g^s$ for all $s \in S$ force z to be at least the cost of every scenario. The minimization of $f(a) + z$ then drives z down to $\max_{s \in S} g^s(a)$. The constraint $\sum_{\tau} a_j^{\tau} \leq 1$ prevents more than one charger type at the same store.

4.3 Instance Preprocessing

Two preprocessing steps are part of the model definition rather than of the solution method. They shape the set of feasible tours and the binary index structure before the solver sees the problem.

4.3.1 Split-Delivery Preprocessing

If a store's demand β_j^s exceeds truck capacity L, one tour cannot serve it fully. To handle this without changing the model structure, preprocessing expands the store into $\lceil \beta_j^s / L \rceil$ pseudo-stores at the same location. Each pseudo-store has demand at most L. The model then treats these pseudo-stores as independent stores.

4.3.2 Fleet Sizing and Multi-Tour Modeling

Two parameters strongly affect model size: $K_{\max}$, the number of available trucks, and $M_{\max}$, the maximum number of tours per truck per day. Both are set tightly to avoid unnecessary indices and binary variables.

$K_{\max}$ is derived from the worst-case total demand across all scenarios, divided by the per-truck daily capacity $L \cdot M_{\max}$, and multiplied by a safety factor of 1.5. The result is capped at $|J|$. $M_{\max}$ is set to 3 by default. This covers most observed multi-tour truck-days while keeping the model tractable.

5 Parameter Calibration

While the mathematical model in Chapter 4 is general, this chapter sets the concrete parameter values for the experiments. The goal is not to copy one exact truck or charger specification, but rather to build a clear planning instance that is realistic enough to make the optimization results useful.

The parameters come from three sources. Some values come from the real historical retailer data, for example store locations, distances, demand scenarios, and unloading-time estimates. Some values come from researching public vehicle and charging data. Other values are planning assumptions, for example the value of waiting time and the lifetime used for spreading out the investment cost over the years. The parameter study in Chapter 8.2 then tests the varying economic assumptions.

5.1 Vehicle and Energy Parameters

The experiments mostly use the Mercedes-Benz eActros 300 as the main vehicle type, with the Volvo FM Electric setup as a secondary option. According to our research, these two models are currently the most widely used electric trucks in the German grocery retail sector. The Mercedes vehicle has a battery capacity of 336 kWh and a payload limit of 17 700 kg. The implementation also includes a Volvo setup with a 360 kWh battery and 16 200 kg payload. The final reported runs use the Mercedes setup so the results are easier to compare. The battery policy is simple. Every truck starts with an initial state of charge of 80 %. The battery may not fall below 20 % at stores or at depot return. This 20 % to 80 % window is chosen because battery discharging behaves mostly linearly within this range. Furthermore, charging up to 80 % can also be modeled as a linear process, whereas charging above this threshold is heavily throttled by the vehicle's battery management system. Limiting the maximum charge to 80 % and preventing deep discharges below 20 % also significantly reduces battery degradation and promotes long-term battery longevity [26]. This follows the common electric-vehicle planning idea that not every nominal kWh should be treated as usable energy for driving [27, 15].

Table 5.1: Main physical parameters used in the computational study.

Parameter	Value	Role in the model
Battery capacity	336 kWh	available truck energy
Payload limit	17 700 kg	maximum delivered load per tour
Initial SoC	80 %	start level for each truck
Minimum SoC	20 %	safety reserve at stores and depot return
Vehicle type	Mercedes eActros 300	typical electric grocery truck

5.2 Cost Parameters

The objective contains charger investment cost and daily operating cost. The operating cost includes electricity, waiting time, warehouse charging, overnight charging, and a fixed truck-use cost. All costs are expressed in EUR.

Table 5.2: Baseline economic parameters.

Parameter	Baseline value	Interpretation
Electricity cost d	0.25 EUR/kWh	Commercial charging price used in the main runs.
Waiting cost h	50 EUR/h	Planning value for truck and driver idle time during charging. It is intentionally higher than a pure wage estimate.
Truck fixed cost F	254 EUR/day	Yearly truck cost broken down to a daily value, plus the cost of one driver day for the employer.
Charger lifetime	10 years	Lifetime used to divide the charger cost evenly across the years.
Operating days	290 days/year	Approximate six-day operation.

While the waiting cost is not a measured wage, it is a planning value that reflects the fact that a truck standing at a charger cannot serve another tour, and that long charging stops make the daily plan less robust. This is why it is varied in the sensitivity study.

Fixed truck cost. The fixed truck-use cost is calculated as

$$F_v = \frac{p_v}{n^{\text{truck}} D_{\text{year}}} + c_{\text{day}}^{\text{driver}}.$$

Here p_v is a typical vehicle price, n^{truck} is the truck lifetime in years, D_{year} is the number of operating days per year, and $c_{\text{day}}^{\text{driver}}$ is one employer-side driver day. The thesis uses a truck lifetime of 9 years, based on the German tax depreciation standard [28]. The driver cost is based on typical German salary benchmarks. The vehicle price is estimated from current European market listings. This gives a Mercedes fixed cost of about 254 EUR per operating

day.

5.3 Charging Infrastructure

The store charger set contains six charger types. This allows the model to choose between cheap slow charging and expensive fast charging during the experiments.

Table 5.3: Store charger types used in the final experiments.

Type	Rated power	Installation cost
AC	22 kW	7,500 EUR
DC	40 kW	22,500 EUR
DC	50 kW	33,000 EUR
DC	90 kW	60,000 EUR
DC	120 kW	75,000 EUR
DC	150 kW	100,000 EUR

The warehouse charger is fixed at 22 kW AC. This is a deliberate modeling choice, even though a real depot might use faster chargers. Slow overnight charging keeps the peak electricity demand low, which protects the local power grid when many trucks charge at once. Most importantly, keeping the depot charger fixed ensures the model focuses only on the main question: where to install chargers at the stores.

A 22 kW AC charger corresponds to a standard three-phase low-voltage connection [29]. For larger truck depots, grid planning quickly moves toward medium-voltage connections [18], which supports the main idea of the thesis: store charging can reduce pressure on the depot and can use unloading time for opportunity charging.

Table 5.4: Fixed depot parameters.

Parameter	Value
Warehouse charger	fixed 22 kW AC
Overnight charging window	10 h
Warehouse loading time	11.51 min + 0.004184 min/kg
Warehouse loading-time factor	0.6

The sensitivity study varies electricity cost, waiting cost, and charger investment cost. It does not change the physical vehicle model.

6 Solution Algorithm

The full robust CSPP is too large to solve as a single model for the final retailer instance. The solution method simplifies the problem by dividing the stores into several clusters. Each cluster is then solved by starting with a small number of demand scenarios and adding more step by step in a loop. While the output does not mathematically prove global optimality, it is a practical planning method that produces valid charger placements for the entire retailer within a predictable time.

This chapter explains how the complete algorithm determines the optimal charger placements and calculates the total costs for each of the given scenarios using a concrete illustrative example. The concrete example is cluster 6 taken from the main full-instance run from the real grocery retailer dataset, using 24 stores as target, 20 stores as minimum and 28 stores as maximum per cluster. Because this is the hardest cluster in that run, it contains 28 stores and required about 21.96 wall-clock hours to solve the first stage, responsible for calculating the optimal charger placements. Section 6.6 follows this same cluster through the full pipeline.

6.1 Model Complexity

The main complexity of the model lies in the second stage of the model. For each scenario, the model contains routing, charging, waiting, battery, and truck-use variables. Since these variables depend on the number of stores, trucks, and routes, the total number of variables for n stores, K trucks, and M tours is roughly

$$N_{\text{var}}^{(2)} \approx 2n(n+1)KM.$$

This total increases very quickly as more stores are added to the problem. The robust model repeats this structure for every single scenario, which is why the full 259-store problem cannot be solved simply by giving the computer more time.

Large Mixed-Integer Programming (MIP) gaps must be interpreted carefully. The solver often finds good solutions early in the process, even when the gap remains high because the lower bound cannot be significantly increased. Because of this, the experiments focus on how the solution improves over time and how long each stage takes, rather than looking only at the final results.

6.2 Clustering

Clustering reduces the number of stores the solver must handle at the same time. Instead of solving one large robust CSPP for all stores, the algorithm solves a smaller model for each cluster. This is an approximation, because trucks are not allowed to drive from one cluster to another. It is still a necessary step, since the full problem cannot be solved in a reasonable time.

The thesis uses four clustering methods:

- *geographic* clustering, which groups stores that are close to each other on the map, using k-means as the main step;
- *angular slices (store-count)*, which cuts the area around the depot into pie slices with the same number of stores in each slice;
- *angular slices (demand-aware)*, which cuts the same kind of slices, but balances the total demand instead of the store count;
- *tour containment*, which tries to keep each historical delivery tour inside a single cluster.

All four methods build the same kind of arc set in the end. The depot is connected to every store in both directions. An arc between two stores is only kept if both stores belong to the same cluster. With cluster assignment $C(j)$, the arc set is

$$A_{\text{cluster}} = \{(0, j), (j, 0) : j \in J\} \cup \{(i, j) : i, j \in J,\ i \neq j,\ C(i) = C(j)\}.$$

The full arc set is also stored for comparison, but the clustered CSPP stages use A_{cluster}. This is the step that makes the model smaller: store-to-store arcs across clusters are removed, while every store can still be reached from the depot.

6.2.1 Geographic clustering

The geographic method first runs k-means on the latitude and longitude of the stores. The depot is included when k-means is fitted, but its label is dropped afterwards. Only the store labels are kept. Including the depot makes the first cluster centers shift slightly toward it, which gives a better starting point than fitting on stores alone.

The first k-means labels are then improved by a small assignment MIP. Let J be the set of stores, K the number of clusters, and $d_{j,c}$ the straight-line distance from store j to k-means

center c. The reassignment model is

$$\min \quad \sum_{j \in J} \sum_{c=1}^{K} d_{j,c} x_{j,c} \tag{6.1}$$

$$\text{s.t.} \quad \sum_{c=1}^{K} x_{j,c} = 1 \qquad \forall j \in J, \tag{6.2}$$

$$n_{\min} \leq \sum_{j \in J} x_{j,c} \leq n_{\max} \qquad \forall c \in \{1, \ldots, K\}, \tag{6.3}$$

$$x_{j,c} \in \{0, 1\} \qquad \forall j \in J,\ c \in \{1, \ldots, K\}. \tag{6.4}$$

The number of clusters K is set from the target store count and the upper bound $n_{\max}$, so that every cluster can stay within the size limits. If the reassignment model has no feasible solution, the algorithm keeps the original k-means labels. The geographic method is therefore a distance-based method with hard size limits.

6.2.2 Angular slice clustering

The angular methods place the depot at the center and sort the stores by the angle at which they appear around the depot. This gives a circular order of stores from 0 to 2π.

Before the stores are split into clusters, the algorithm looks for the largest empty gap in this circular order, that is, the place on the circle where two neighboring stores are the furthest apart in angle. The cut between the first and the last cluster is placed inside this gap. The reason is simple: angles wrap around (2π is the same as 0), so a fixed cut at angle 0 could fall in the middle of a tight group of stores and split it for no good reason. Cutting in the largest gap avoids this and makes sure that the boundary between two clusters lies in an area with few stores.

In the store-count variant, the sorted stores are then split into K connected groups with almost the same number of stores in each group. Each cluster covers one slice around the depot, and neighboring clusters cover neighboring slices.

The demand-aware variant uses the same circular order, but it does not split by count alone. Instead, it uses dynamic programming. Let the sorted stores be $j_1, \ldots, j_n$, and let q_j be the largest demand of store j across all scenarios. The dynamic program looks for connected groups that respect the size limits and keep the total demand of each cluster as close as possible to the target demand $\bar{q} = \sum_{j \in J} q_j / K$:

$$\min \sum_{c=1}^{K} \left(\sum_{j \in G_c} q_j - \bar{q} \right)^2,$$

where $G_1, \ldots, G_K$ are connected parts of the sorted order and $n_{\min} \leq |G_c| \leq n_{\max}$. If no feasible split exists under the size limits, the algorithm falls back to the equal-count split. This method keeps the same pie-slice shape, but tries to avoid clusters that have a much higher worst-case demand than others.

6.2.3 Tour-containment clustering

Tour-containment clustering uses historical delivery tours instead of coordinates or demand. The raw data is grouped by tour name and matched to the internal store indices. Repeated stops within the same tour are removed, and tours with fewer than two stores are dropped. Tours that visit the same set of stores are merged into one pattern. The number of original tours that share the pattern becomes its weight, so a tour pattern that appears often counts more than a pattern that appears only once.

Let J be the set of stores, K the number of clusters, and $\mathcal{T}$ the set of merged tour patterns that are small enough to fit into one cluster:

$$\mathcal{T} = \{t \subseteq J : |t| \geq 2,\ |t| \leq n_{\max}\}.$$

Each tour $t \in \mathcal{T}$ has weight w_t, equal to how often the pattern appears after merging. The model is

$$\begin{aligned}
\max \quad & \sum_{t \in \mathcal{T}} \sum_{c=1}^{K} w_t v_{t,c} && && \text{(TOUR-CONTAINMENT)} \\
\text{s.t.} \quad & \sum_{c=1}^{K} x_{j,c} = 1 && \forall j \in J, && (6.5) \\
& n_{\min} \leq \sum_{j \in J} x_{j,c} \leq n_{\max} && \forall c \in \{1, \ldots, K\}, && (6.6) \\
& v_{t,c} \leq \frac{1}{|t|} \sum_{j \in t} x_{j,c} && \forall t \in \mathcal{T},\ c \in \{1, \ldots, K\}, && (6.7) \\
& x_{j,c} \in \{0, 1\} && \forall j \in J,\ c \in \{1, \ldots, K\}, && (6.8) \\
& v_{t,c} \in \{0, 1\} && \forall t \in \mathcal{T},\ c \in \{1, \ldots, K\}. && (6.9)
\end{aligned}$$

The variable $x_{j,c} = 1$ means that store j is placed in cluster c, and $v_{t,c} = 1$ means that the full tour pattern t is inside cluster c. The linking constraint has the same effect as a separate constraint $v_{t,c} \leq x_{j,c}$ for each store j in t: since $v_{t,c}$ is binary, it can only be one when every store of the tour is in the same cluster. Using one combined constraint per tour and cluster keeps the model smaller than writing one constraint per store, tour, and cluster. Tour patterns larger than $n_{\max}$ are removed before solving, because they cannot fit into any cluster.

Unlike the angular and geographic methods, this model does not aim directly at compact regions. The objective rewards keeping recurring tours complete inside a single cluster. Compactness only enters through the original tour shapes and the cluster size limits. After the assignment is found, the same depot-plus-within-cluster arc set is built.

The comparison in Chapter 8.1 shows that demand-aware angular slices give the best results, so the final runs in Chapter 8 use this method. In the example below, the stores are split into 11 clusters of about 24 stores each, and we follow cluster 6 through the algorithm.

6.3 Stage 1: Scenario Addition with Bound-Based Pruning

6.3.1 Algorithmic idea

Each cluster has 31 historical demand scenarios available. Adding all of them at once as separate constraint blocks in one big robust MIP would make the model too complex to solve. Instead, Stage 1 follows the approximate scenario bracketing procedure for two-stage robust mixed-integer problems [21] developed by the TUM Chair of Optimization and Sustainable Decision Making. This method only looks at a small group of active scenarios, called $D \subseteq S$, at a time. At each outer step, the algorithm solves the main model (the restricted master problem) using only the scenarios in D. The remaining scenarios in $S \setminus D$ are then checked using short heuristic mip instances. Only scenarios that can still change the worst case are added to D in the next step. This way, the algorithm focuses on the most critical scenarios and avoids spending time on those that are clearly not the worst case.

The main loop a cluster works as follows:

1. Solve the main model using only the scenarios in D to find a charger plan.
2. run heuristic and bound updates on every scenario in $S \setminus D$ to maintain a lower bound $\mathrm{LB}[k]$ and an upper bound $\mathrm{UB}[k]$.
3. Remove scenarios that are clearly not the worst case. This happens if their maximum cost is already lower than what the main model expects or if they are less expensive than another scenario.
4. Find the most difficult remaining scenario and add it to the active set D.
5. Stop the loop when all scenarios have been handled, the target error margin is reached, or the time limit is up.

For a master solution with first-stage cost $\tilde{f}$, robust objective $\tilde{z}$, reported relative MIP gap γ,

and target gap p, the z-bound used from [21] is

$$z_{\text{bound}} = \frac{1-p}{1-\gamma}\,\tilde{z} + \frac{\gamma-p}{1-\gamma}\,\tilde{f}. \tag{6.10}$$

Two rules use this bound to remove unnecessary scenarios:

- **z-bound rule.** Drop scenario k if its highest possible cost $\mathrm{UB}[k]$ is below the z-bound. This scenario cannot be more expensive than the current worst case that the solver is already okay with, so adding it would not change the plan.
- **Dominance rule.** Drop scenario k if its highest cost is below the lowest possible cost of another scenario. Another scenario is guaranteed to be more expensive, so scenario k can never be the worst case.

If every unselected scenario is either added to D or removed by these rules, Stage 1 stops.

6.3.2 Example: Cluster 6

The first outer step starts with no scenario in D, and runs heuristics on all 31 scenarios to select the initial worst scenario. After this step, every unselected scenario has an initial upper bound of about 2032 EUR (from the master second-stage relaxation), while only a few have a tight lower bound because most second-stage probes have not finished yet. The most critical scenario is added to D, and the master is solved again.

Over the next steps, D grows and the upper bounds for individual scenarios are improved by repeated heuristic and second-stage probes. Whenever a candidate's upper bound drops below the master's used z-bound, it is permanently removed. For Cluster 6, this happens to four scenarios. Their final upper bounds are between 302 and 306 EUR, while the master's used bound at the end is $z_{\text{bound}} \approx 328$ EUR. None of these four scenarios is added to D, because doing so cannot improve the robust decision under the target gap.

After 28 outer steps, the cluster 6 solve stops with 27 active scenarios in D_6. Only four scenarios, $\{9, 16, 23, 30\}$, are excluded by the z-bound rule. The reached relative MIP gap is 55.4 %. The first-stage decision installs 15 store chargers, which creates a first-stage charger investment cost share of $\tilde{f} \approx 228.79$ EUR per day; the master's worst-case second-stage part is bounded above by $\tilde{z} - \tilde{f} \approx 2632$ EUR.

This example shows two important points. First, the loop finds a good charger plan very early, and much of the remaining time is spent proving that no other scenarios will change the result. Second, a high reported error margin (gap) does not necessarily indicate a poor plan, as it is often heavily inflated by the difficulty of obtaining tight bounds for the unselected scenarios. These findings support the decision in Chapter 7 to limit the time for each step instead of running one very long calculation.

6.4 Stage 2: Scenario Evaluation

With the charger plan fixed from Stage 1, Stage 2 calculates the optimal routes and charging schedules for every demand scenario—including those omitted from the initial placement.

The algorithm computes the operating cost g_C^s for each cluster C and scenario s. Summing these costs across all clusters yields the total cost for each scenario:

$$g_{\text{sum}}^s = \sum_C g_C^s.$$

The scenario with the highest total cost is the global worst case. In our example, Stage 2 does not change the charger plan for cluster 6, but it calculates the costs needed for Stage 3 to identify which clusters contribute most to the total cost.

The calculations in Stage 2 are much faster than in Stage 1 because each scenario is handled separately. Most of these tests finish in just a few seconds.

6.5 Stage 3: Reoptimization

Stage 3 is the final step to ensure the plan works well for all stores combined. In Stage 1, a charger plan is chosen for each cluster separately. However, a scenario that seems unimportant for a single cluster can still become a problem when we look at all clusters together. Stage 3 identifies these scenarios and re-solves the affected clusters with the new information.

6.5.1 Loop structure

Let A be the global active set of difficult scenarios collected across Stage 3 steps, starting as empty ($A = \emptyset$). Each reoptimization step performs the following actions.

Step a (`eval_before`). Test the current plan against all scenarios and find the most expensive one ($s^\star$). In the first iteration, this step simply uses the evaluation results already calculated in Stage 2.

Step b (convergence check). Decide whether to stop. The loop ends if we have already handled this scenario or if all clusters already consider it in their plans.

Step c (expand A). Add this scenario to the difficult group A.

Step d (`cluster_solve`). Re-solve any clusters that did not originally include this scenario. This produces a new candidate plan.

Step e (`eval_after`). Test the new candidate plan. To save time, we only re-calculate the costs for the clusters that actually changed.

Step f (accept or reject). Only accept the new plan if it does not increase the total cost. If the new plan is better, it becomes the current plan. If it is worse, we keep the previous plan but continue searching if the worst scenario has changed.

Termination. The loop stops when it can no longer improve the result or after a set number of steps. The best plan found is the final output. This rule ensures that Stage 3 will never make the total cost worse.

6.5.2 Reoptimization Behavior in the Example

In the example with 24 stores per cluster, the most expensive scenario was scenario 27. Since this scenario was already included in the active sets of 10 out of the 11 clusters, Stage 3 only had to re-solve one cluster. That cluster's plan did not change for that cluster, so Stage 3 stopped quickly. This shows that Stage 1 already did a good job of identifying the most important scenarios.

In contrast, the experiments with 12 stores per cluster in Chapter 8 show that smaller clusters often miss important scenarios. In that case, Stage 3 runs for several steps to improve the results. The rule for accepting only better plans ensures that Stage 3 only makes useful corrections without ruining a good solution.

6.6 Pipeline Summary

In summary, the full process for finding a charger plan follows these steps:

1. **Cluster the stores.** Divide all stores into smaller groups using the chosen clustering method.
2. **Stage 1.** For each cluster, find a charger plan and a group of important scenarios using the loop described earlier.
3. **Stage 2.** Fix the plans for all clusters and calculate the operating costs for every scenario. This helps identify the global worst case.
4. **Stage 3.** Re-solve clusters that missed the global worst case. Accept new plans only if they improve the overall result.

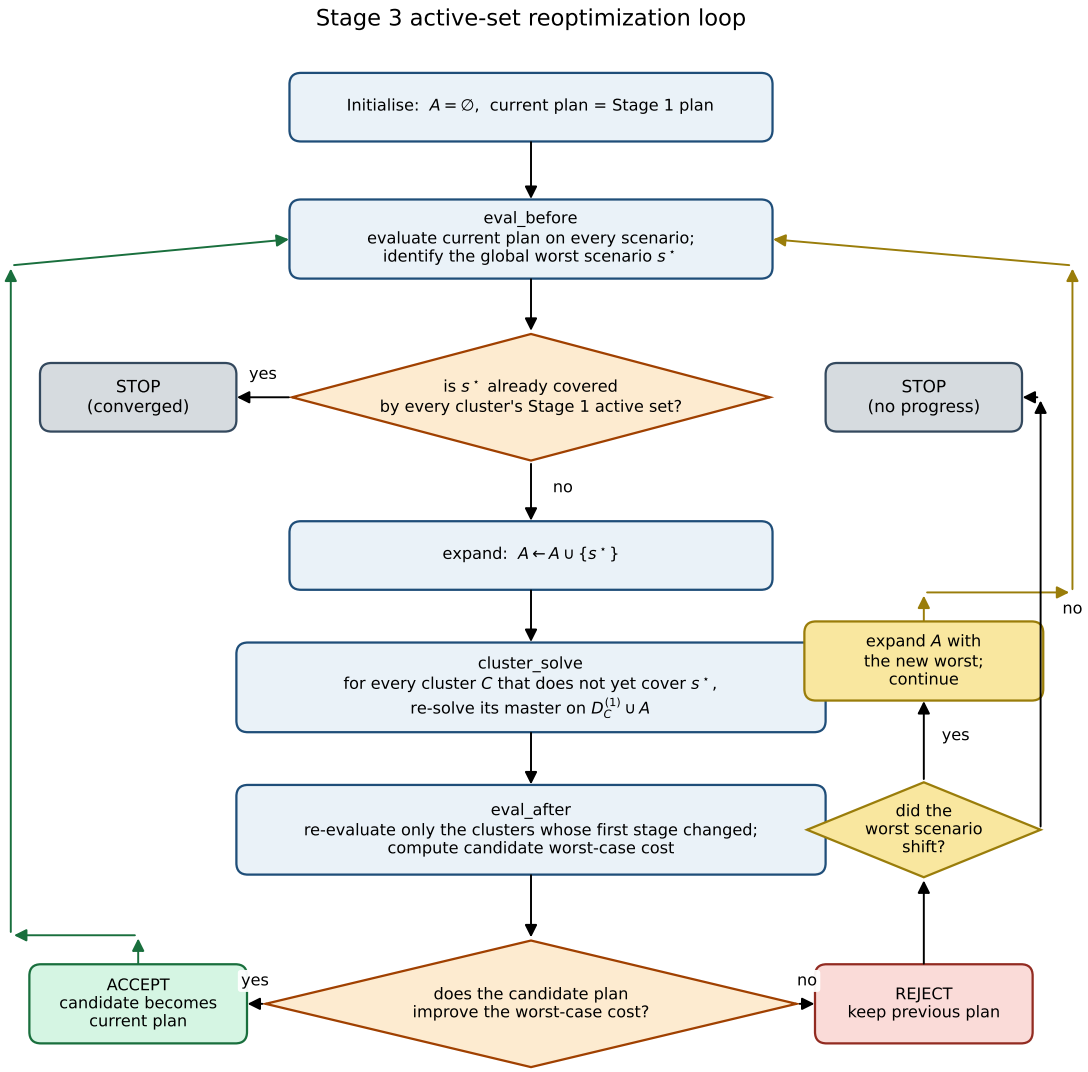

Figure 6.1: Stage 3 active-set reoptimization loop. The candidate plan is accepted only if it does not increase the total cost. If the plan is rejected but the worst scenario has changed, the loop continues with a new scenario in A.

The example of cluster 6 shows how this works in practice. In Stage 1, the algorithm finds a plan and removes scenarios that are not important. In Stage 2, the plan is tested against all scenarios, and scenario 27 is found to be the most expensive. Since scenario 27 was already handled during Stage 1 for almost all clusters, Stage 3 does not need to make major changes. This leads to the final result where cluster 6 is part of a plan with a total daily cost of about EUR 17,379 (Chapter 8).

7 Runtime Calibration

The algorithm described in Chapter 6 depends on several MIP time limits. We need to decide how long the computer should spend on each iteration of Stage 1 and each evaluation in Stage 2. Setting these limits correctly is important because the entire process takes many hours. If a limit is too high, the computer is wasting time on a problem whose objective value is not improving . If it is too low, the computer might stop before finding a good result.

This chapter explains the tests used to find the suiting time limits for the CSPP. We look at how different cluster sizes affect the time needed for Stage 1 charger placement and how quickly Stage 2 reaches a stable solution for the daily operating costs. Finally, we describe a "bucket policy" that automatically sets the time limits based on the difficulty of each cluster.

7.1 Testing Stage 1 Cluster Sizes

7.1.1 Setup

We tested Stage 1 using target cluster sizes of 12, 18, and 24 stores. For each test, we let the algorithm run until it finished. In these tests, we allowed every cluster to finish completely to observe how cluster size affects computational difficulty.

7.1.2 Results

All 13 test clusters across the three targets finished successfully. The results show a steep increase in required runtime as clusters grow larger. On average, Stage 1 took 39.3 min per cluster for the 12-store target (6 clusters), 56.5 min for the 18-store target (4 clusters), and jumped to 163.6 min for the 24-store target (3 clusters). This exponential scaling confirms that while smaller clusters can be solved quickly, larger clusters required for the final runs will demand strict time limits to remain computationally feasible within a realistic overall time budget.

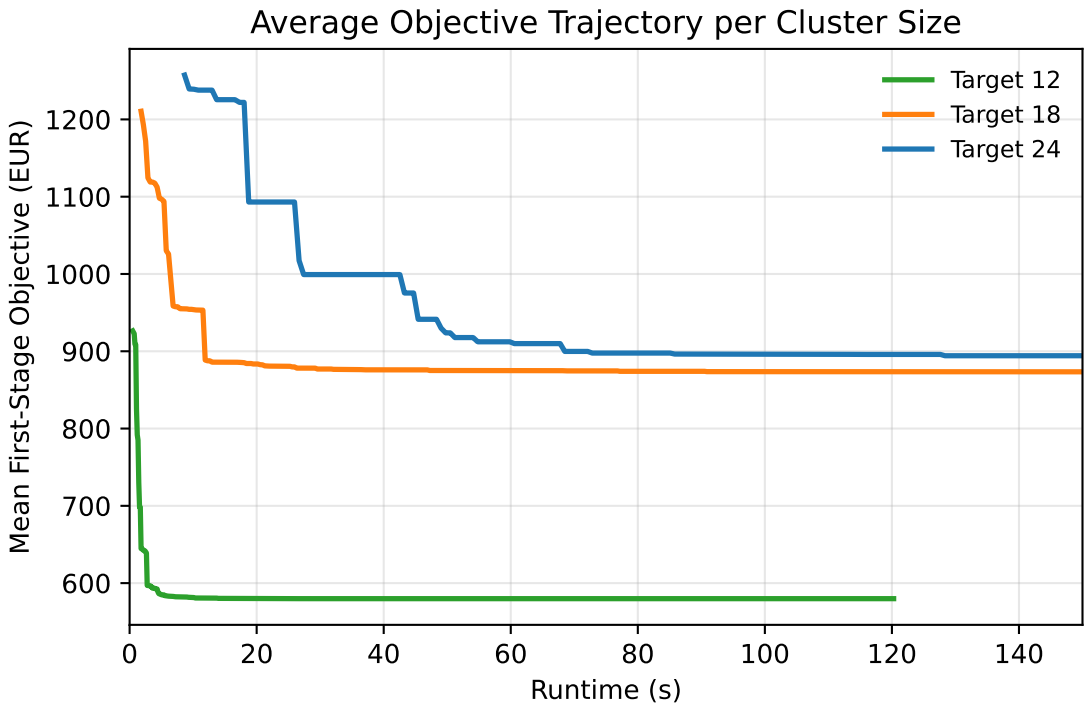

Figure 7.1: Stage 1 objective trajectory for different target cluster sizes. The progress slows down significantly and reaches a plateau early, while the remaining time only provides marginal improvements to the final incumbent.

7.2 Stage 2 Plateau Probe

7.2.1 Setup

For Stage 2, we used the same target cluster-size runs as in the Stage 1 sweep. The live solver logs from the `ct12`, `ct18`, and `ct24` runs were grouped by cluster and scenario. For each group, we tracked the best incumbent found over time and then averaged these trajectories within each target cluster size.

7.2.2 Results

The results show that Stage 2 evaluations are also strongly affected by cluster size. For the 12-store target, the median evaluation was within one percent of its final value after about 35 s. For the 18-store target, the median was about 49 s. For the 24-store target, however, the median increased to about 184 s, and some evaluations were still improving close to the nominal time limit.

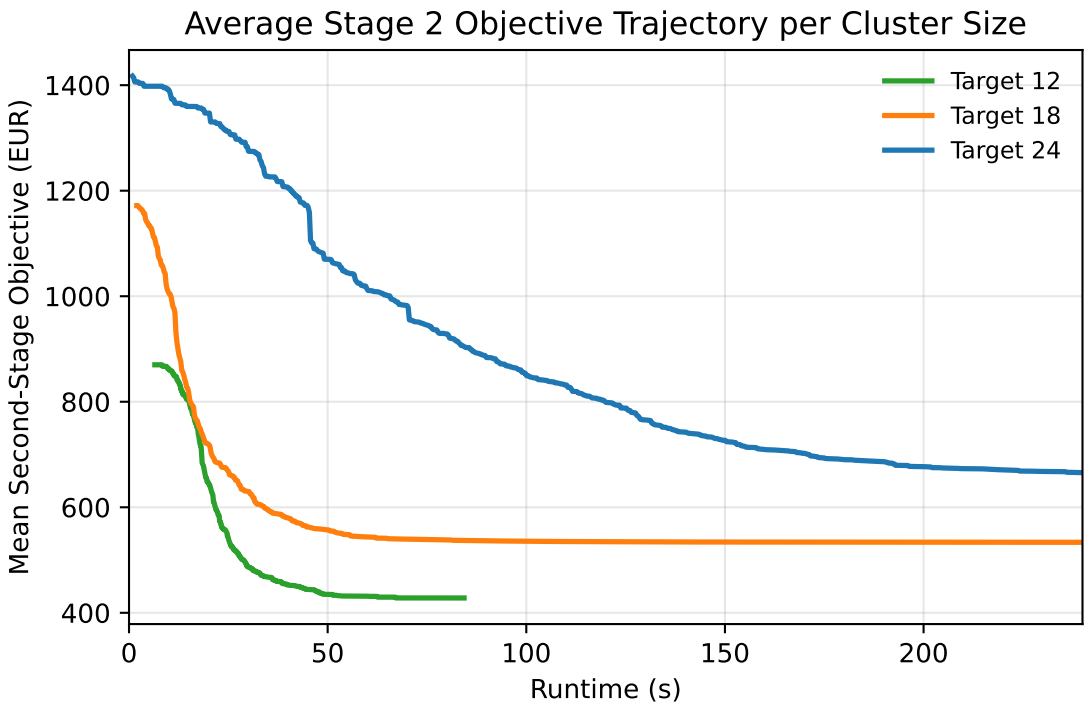

Figure 7.2: Stage 2 objective trajectory for the same target cluster sizes as the Stage 1 sweep. Small and medium clusters stabilize quickly, while the 24-store target continues to benefit from additional time.

This pattern supports bucket-dependent Stage 2 limits rather than a single long limit for

every evaluation. Small and medium evaluations would waste time under a uniformly large limit, while the continuation rule gives larger or still-improving evaluations additional rounds when needed.

7.3 Number of MIP Solves

The reason for calibrating the time limits carefully is that the decomposition creates many individual MIP solves. Let C be the number of clusters, S the number of demand scenarios, and I_3 the number of Stage 3 reoptimization iterations that are evaluated. The workload is calculated as follows:

- Stage 1 solves one clustered first-stage master problem for each cluster and scenario-addition step. Since the implementation can add up to three scenarios per step, the planning count is $C \cdot \lceil S/3 \rceil$.
- Stage 2 fixes the charger plan and evaluates each cluster under each scenario. This gives $C \cdot S$ second-stage operating models.
- Stage 3 reoptimizes selected first-stage decisions and evaluates candidate plans. In the cluster-evaluation variant used for the full-instance runs, one iteration requires up to $C + 2CS$ models: one reoptimization model per cluster, one evaluation of the plan before the change, and one evaluation after the change.

Table 7.1 applies these formulas to the two full-instance runs. Even before accounting for continuation rounds, the 12-store run requires hundreds of Stage 2 models and thousands of potential Stage 3 models. The 24-store run did not require a successful reoptimization step; the Stage 3 number shown for this run is only the first reoptimization attempt before the algorithm stops. This is why the time limits must be short enough to avoid wasting time on stalled solves but flexible enough to continue unexpectedly hard evaluations. The counts are planning counts for the model workload; the implementation can reduce the realized number in Stage 3 through cached evaluations and by skipping clusters that are already covered by the active scenario set.

7.4 Bucket Policy

The test results show that using the same time limit for every cluster is not the best approach. Small clusters need less time than large or difficult ones. To handle this, the algorithm uses "runtime buckets." Before a cluster is solved, it is assigned to a bucket based on its size,

Table 7.1: Planned number of MIP solves for the two full-instance runs. The Stage 3 count for the 24-store run is only the first reoptimization attempt, after which the algorithm stops because no improvement is needed.

Run	C	S	I_3	Stage 1	Stage 2	Stage 3
12 stores per cluster	22	31	3	242	682	4158
24 stores per cluster	11	31	1	121	341	693

distance from the depot, and the average number of active stores. Each bucket has its own standard time limit for Stage 1 and Stage 2. If the solver is still making progress when the time runs out, it can be given extra time.

Table 7.2: Shared runtime bucket definition.

Bucket	Definition
`xsmall`	stores ≤ 12
`small`	stores ≤ 14
`medium`	stores ≤ 16, avg_depot_km < 15, mean_active_customers < 8.5
`hard`	stores ≤ 20, avg_depot_km < 30, mean_active_customers < 10
`xhard`	all remaining cases

Table 7.3: Nominal time limits used by the shared bucket policy.

Bucket	Stage 1 [s]	Stage 2 [s]
`xsmall`	120	45
`small`	180	60
`medium`	240	120
`hard`	300	180
`xhard`	360	240

The values in Table 7.3 are base time limits rather than absolute hard caps. The Stage 1 limit applies to each iteration of the main model, while the Stage 2 limit applies to each evaluation. If a solve reaches its time limit without an incumbent, or if the incumbent was still improving late in the round, the implementation allows up to three continuation rounds. A late improvement means that the last incumbent improvement occurred after 80% of the current round. Each continuation increases the cumulative time budget by one additional base limit. Thus, an `xhard` Stage 1 master solve starts with 360 s but can receive up to 1440 s in total; an `xhard` Stage 2 evaluation starts with 240 s and can receive up to 960 s. Stage 3 uses the same Stage 1 continuation for cluster re-solves and the same Stage 2 continuation for candidate evaluations.

This continuation rule was implemented as a safeguard for unexpectedly hard evaluations.

It avoids using overly conservative base limits for every solve, while still allowing the algorithm to spend more time on cases where no incumbent has been found or where the solver is still making late progress.

7.5 Time Limits in the Final Experiments

In the main run with 24 stores per cluster, all 11 clusters are classified as "extra hard" (xhard) because they are large. In the run with 12 stores per cluster, the clusters fall into various smaller buckets. The results in Chapter 8 show that these time limits work well. Good charger plans are found early, and the remaining time is used to confirm those results. This confirms that the time limits are doing exactly what we intended.

8 Results

This chapter presents all experimental results. We first compare four clustering methods on a smaller 96-store instance to select the method used in the full experiments (Section 8.1). We then explore how cost parameters influence charger placement on a single cluster from the same 96-store instance (Section 8.2). Section 8.3 describes the setup for the full-scale experiments, and Section 8.4 reports the final results on the 259-store instance.

8.1 Clustering Method Comparison

Choosing the right clustering method is a very important decision. The way stores are grouped into clusters determines which routes the trucks can take. Since trucks cannot travel between clusters, a bad choice at this stage cannot be fixed later. This section compares four different clustering methods using a smaller instance. Based on the results, we choose the method with the best results for the final, full-scale experiments.

8.1.1 Setup

We tested the clustering methods using a smaller problem with 96 stores and 8 demand scenarios. We split the stores into clusters of about 12 stores each, resulting in about 8 clusters for each method. This size is large enough to see how the methods perform differently but small enough to finish the tests in reasonable time. We compared four methods: geographic clustering, tour containment, demand-aware angular slices, store-count angular slices,. All tests used the same settings and the same algorithm described in earlier chapters. Our goal was to find which method resulted in the lowest total cost for the worst-case scenario.

8.1.2 Results

Figure 8.1 reports the cluster-size distribution produced by each method, and Table 8.1 summarises the cost outcomes after Stage 2 and reoptimization.

Figure 8.1: Cluster-size distribution in the 96-store clustering comparison. Store-count angular slices produce exactly balanced clusters, but this does not produce the best robust result.

Table 8.1: 96-store clustering comparison. Costs are scenario operating costs in EUR. The final worst-case cost is the robust objective after Stage 3.

Method	Cluster sizes	Avg. cost	Stage 2 worst	Worst after reopt.	Chargers
Angular slices (demand-aware)	14,13,16,16,9,10,10,8	3565.20	4620.44	4027.17	32
Geographic	9,15,16,16,11,15,1,13	3413.49	4619.03	4209.18	24
Tour containment	8,16,15,7,15,4,16,15	3247.24	4338.11	4338.11	29
Angular slices (store-count)	12,12,12,12,12,12,12,12	3745.26	4984.62	4668.28	21

8.1.3 Interpretation

Demand-aware angular slices perform best. Initially, the tour containment and geographic clustering methods seemed to perform better. However, after the reoptimization stage of the algorithm, the results changed significantly. The demand-aware angular slices method reached the lowest total cost, making it the best choice. This final result is what matters most since we use all three stages of the algorithm.

Equal cluster sizes are not the most important factor. The original angular-slices method created eight clusters with exactly 12 stores each. Although this seems like a fair split, it actually resulted in the highest cost. This shows that having an equal number of stores in each cluster is less important than factors like demand, distance, and route structure. Two effects explain this. First, equal counts ignore demand imbalance: a cluster of 12 high-volume stores piles far more load onto a few clusters that then dominate the worst case. Second, forcing exactly 12 stores per slice stretches angular boundaries across density gaps, giving sparse regions wide slices that inflate travel and waiting time.

Geographic clustering is risky because of extreme clusters. While geographic clustering achieved a good final cost, it created one cluster that only contained a single store. This is a major risk for a general planning tool, because very small clusters can hide capacity decisions that should really be shared with nearby stores. Such artificially small regions do not reflect the real regional delivery structure. In previous test runs, geographic clustering also produced dense clusters located very far away from the warehouse, which made them infeasible because the trucks could not reach all stores within their range.

Tour containment is strong on average but not best for robustness. The tour containment method delivered the lowest average cost, but after the reoptimization step, it was no longer the best robust solution. While keeping historical tours together is useful, it can still miss a better grouping when future worst-case days look different from the historical tour shapes.

Reoptimization significantly improved the results for most methods. It improved the final worst case for demand-aware angular slices by EUR 593.27, for geographic clustering by EUR 409.85, and for store-count angular slices by EUR 316.34. These numbers show that the correction step can completely change which method is considered the best.

8.1.4 Choice for the Final Experiments

For our final experiments in Section 8.4, we chose the demand-aware angular slices method. It produced the best final robust result, avoids creating extreme one-store clusters, and does not rely on complex tour-containment preprocessing. It also fits the depot-centered layout of the retailer perfectly, naturally splitting stores by their angle around the warehouse while balancing their demand.

Our tests used 12 stores per cluster, but the final experiments use both 12 and 24 stores. While using 24 stores per cluster takes longer to calculate, it provides a more robust plan. Further comparisons between these methods on a larger scale are left for future research, as noted in Chapter 9.

8.2 Cost Sensitivity Study

In Chapter 5, we set initial costs for electricity, driver waiting time, and charging equipment. This section explores how changing these costs affects the decision of where to place chargers. The goal is to observe how the model behaves as real-world cost values change, demonstrating that the model adapts to different economic conditions.

8.2.1 Setup

We conducted these tests on a single cluster of 16 stores. Using one cluster allows us to run many tests quickly while still getting useful results. For each test, we changed one specific cost while keeping the others the same. We focused on electricity costs, waiting costs, and the cost of the chargers themselves. We only looked at the initial decision of where to place chargers, skipping the later stages of the algorithm.

8.2.2 Results

Waiting cost. The number of installed chargers goes up as the waiting cost increases. For $h = 25, 37.5, 50, 75$, and 100 EUR/h, the number of installed chargers is 5, 7, 8, 9, and 9. This is expected, because when driver waiting time is more expensive, having fast charging capacity becomes much more valuable to avoid delays. Eventually, the curve flattens out, because once enough capacity is installed, making waiting even more expensive does not create any new useful locations for a charger.

Charger investment cost. The charger-cost multiplier has the strongest effect overall. For $\chi = 0.5, 0.75, 1.0, 1.5$, and 2.0, the number of installed chargers is 9, 10, 8, 6, and 2. The general

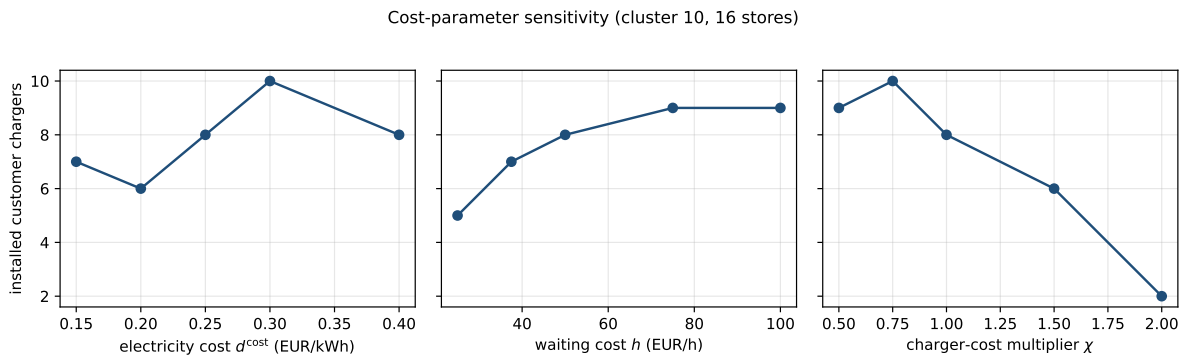

Figure 8.2: Number of installed store chargers as different costs are changed.

trend goes down, meaning that fewer chargers are installed as they become more expensive. Small variations (like the slight increase from $\chi = 0.5$ to $\chi = 0.75$) are likely due to the remaining mathematical gaps rather than a change in the overall trend.

Electricity cost. The electricity cost does not show a clear or simple trend. For $d^{\text{cost}} = 0.15, 0.20, 0.25, 0.30$, and 0.40 EUR/kWh, the number of installed chargers is 7, 6, 8, 10, and 8. This makes sense because the energy needed for the routes is mostly fixed; changing the electricity price scales the operating cost up or down but does not strongly change where a charger should be placed.

Trend. Waiting costs and charger investment costs shape the final plan much more than electricity costs do in this range.

8.2.3 Key Findings

To get the best results from this model, it is most important to focus on accurate costs for driver waiting time and charging equipment. While electricity prices affect the total cost, they do not change the charger plan as much.

8.3 Full-Instance Experimental Setup

This section describes the setup for our final, full-scale experiments. The decisions about time limits and clustering methods were made based on the tests in earlier chapters. We conducted two main experiments using the same data and settings, with the only difference being the number of stores in each cluster.

8.3.1 Data and Settings

We used actual delivery data from the retailer for our experiments. After cleaning the data, we were left with 259 stores served from a single warehouse. We used 31 real historical days to represent the variations in demand. All runs use the Mercedes-Benz eActros 300 truck.

Table 8.2: Setup for the full-scale experiments.

Item	Setting
Stores	259
Demand scenarios	31 real historical days
Warehouse	One fixed depot
Vehicle	Mercedes-Benz eActros 300
Clustering	Demand-aware angular slices
Comparison run	12 stores per cluster
Main run	24 stores per cluster

8.3.2 Two Cluster-Size Experiments

We compared two different cluster sizes to see how they affect the results.

12 stores per cluster. The stores were split into 22 clusters. This experiment shows if using smaller clusters makes the problem easier to solve and how it affects the final cost.

24 stores per cluster. The stores were split into 11 clusters. Although these larger clusters take more time to solve, they better preserve natural delivery routes better. This experiment provides our main results.

Comparing these two runs helps us understand the trade-off between the speed of the calculation and the quality of the final plan. We present the results for the smaller clusters first in Section 8.4 before moving to the main results with the larger clusters.

8.3.3 Scope and Consistency

These experiments are designed to determine where to place chargers, how well they perform on different days, and what corrections are needed. All other settings, such as clustering methods and costs, were fixed in earlier chapters. This ensures that the only difference between our two main experiments is the cluster size, allowing for a fair comparison.

Our research process was gradual. Earlier runs showed which parts of the algorithm were too slow or which clustering choices were too risky. Those findings then changed how the final runs were set up. For example, the cluster-size sweep helped create the bucket-based

time limits, and the smaller 96-store test decided the clustering method for both experiments. This section presents the final, refined process that we used to get our results.

8.4 Full-Instance Results

This section presents the final results for our 259-store experiments. Following the plan from Section 8.3, we first examine the results from using 12 stores per cluster, which is faster but less effective. Then, we look at our main results from using 24 stores per cluster. Both experiments completed all three stages of the optimization process.

8.4.1 Store Layout and Demand

Figure 8.3 shows the spatial layout of the 259-store instance and the maximum daily demand at each store.

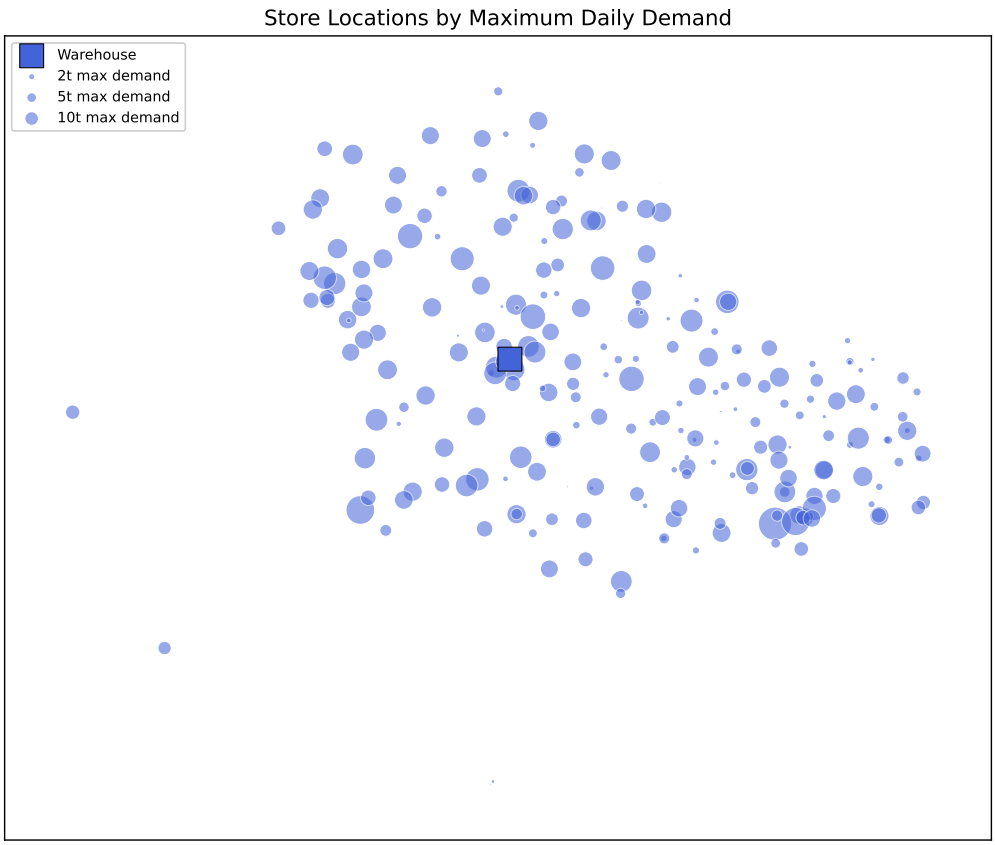

Figure 8.3: Full 259-store instance. Bubble size shows maximum daily store demand.

8.4.2 12 Stores per Cluster

Because the clusters were small, Stage 1 finished very quickly, taking only about 5 hours. This speed is the primary benefit of using smaller clusters, as the entire calculation can be completed in a single day.

Charger placement

This experiment resulted in 94 chargers being installed at a total investment cost of EUR 3 622 000. This plan relied heavily on slower, cheaper chargers compared to our main results, specifically using many more 22 kW AC chargers and fewer medium-power DC chargers.

Total cost before correction

After the second stage evaluated all 31 scenarios, the highest daily operating cost was EUR 23 278.78. This experiment showed that using smaller clusters makes the plan more vulnerable to unexpected demand. Because the clusters are so small, it is harder to adjust delivery routes when one area becomes busy.

Reoptimization trace

During reoptimization, the algorithm performed three iterations to improve the plan.

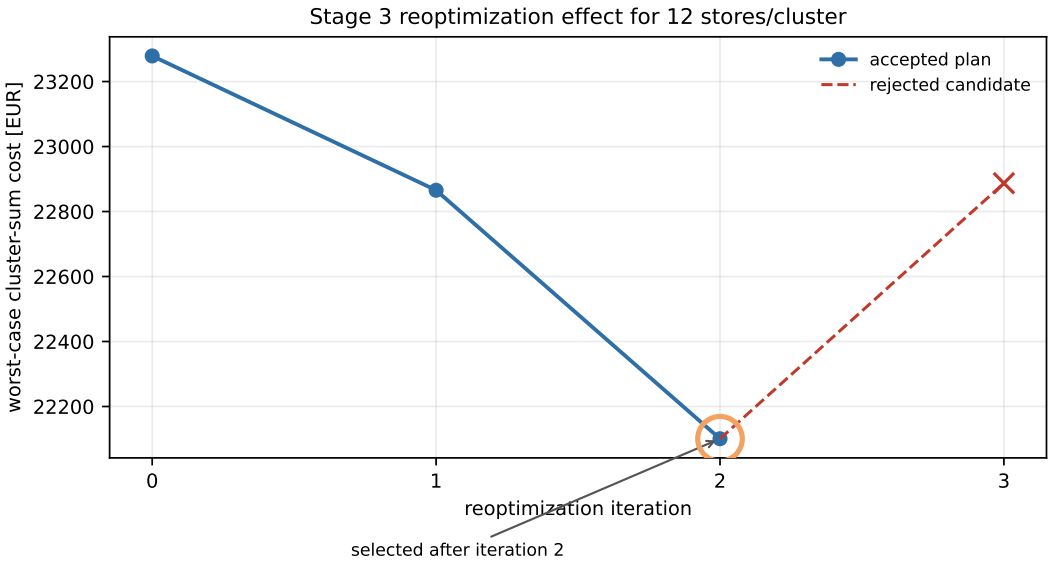

Figure 8.4: Stage 3 reoptimization trace for the 12 stores per cluster run. Iteration 2 is selected because iteration 3 increases the worst-case cost and is rejected.

Iteration 1. Before this step, the worst scenario was scenario 21 at EUR 23 278.78. The set of difficult scenarios A became $\{21\}$. Eleven clusters were solved again, and ten of them changed their placement. This dropped the worst-case cost to EUR 22 865.47, so the candidate was accepted.

Iteration 2. The worst scenario shifted to scenario 13, and the set A grew to $\{13, 21\}$. Fifteen clusters were solved again, and ten changed their placement. The worst-case cost dropped further to EUR 22 100.73. This was also accepted and became the best plan.

Iteration 3. The worst scenario shifted to scenario 20, and A became $\{13, 20, 21\}$. Nine clusters were solved again. However, the new worst scenario (scenario 27) had a cost of EUR 22 887.18, which was worse than the plan from iteration 2. Therefore, the candidate was rejected and the loop stopped.

The final plan from the second step resulted in a daily cost of about 22,101 EUR. This shows that reoptimization successfully improved the plan by 5% and that our rules for accepting only better plans functioned as intended.

Limitations

Although this experiment was fast at first, it required much more time during the final correction stage. More importantly, the final cost was still 27% higher than our main results. This is because small clusters limit the possible delivery routes too much, and no amount of correction can fix a poor initial split. Therefore, we moved on to the more complex experiments with larger clusters.

8.4.3 Main 24-Stores-per-Cluster Run

The 11 clusters in this run range from 20 to 28 stores. Because they are larger and keep more regional routing structure, they are harder to solve, and Stage 1 becomes the biggest computational bottleneck.

Calculation time

When added together, the clusters took 150.46 CPU hours to solve. Because they ran in parallel, the actual wall-clock time was 54.51 hours. The hardest cluster was cluster 6, which had 28 stores and took 21.96 hours across 28 steps to finish with a remaining gap of 55.4 % and 15 installed chargers. The time-limit rules from Chapter 7 managed these long calculations while allowing extra time for improvement.

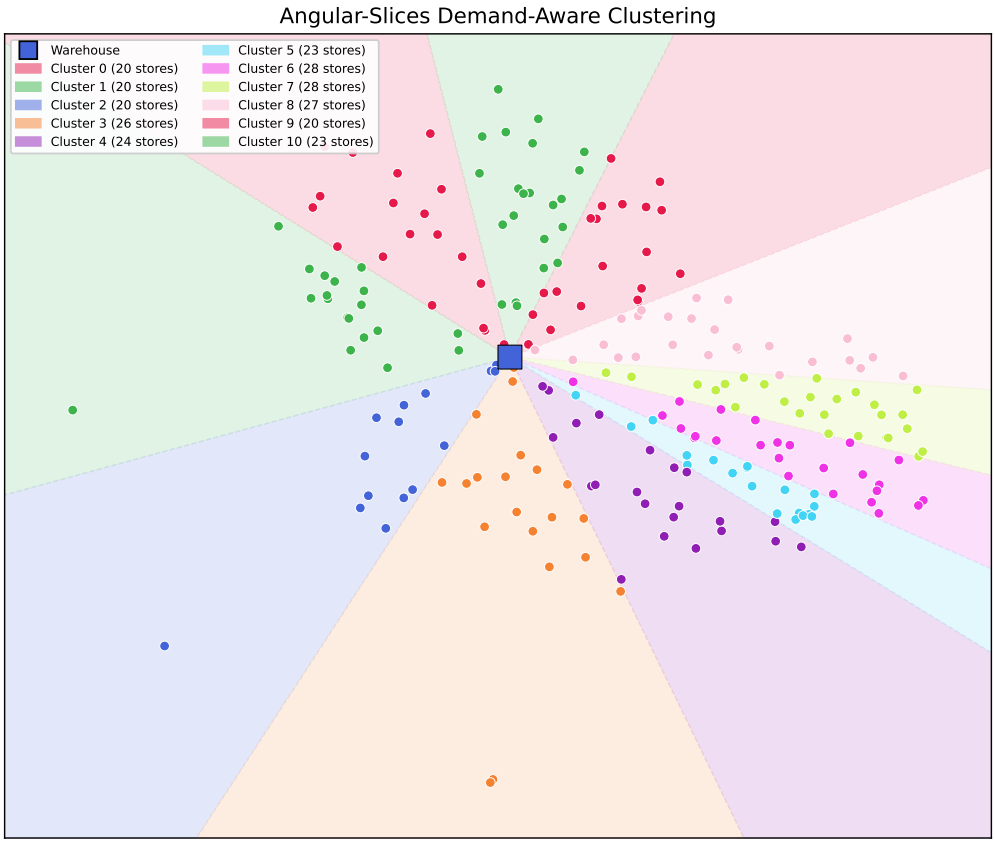

Figure 8.5: Clustering for the main experiment (11 clusters).

Charger placement

This plan installed 95 chargers at a total investment cost of EUR 4 805 000. Unlike the previous experiment, this plan used a wide variety of chargers: 13 chargers at 22 kW AC, 17 at 40 kW DC, 20 at 50 kW DC, 9 at 90 kW DC, 19 at 120 kW DC, and 17 at 150 kW DC. This use of more high-power DC chargers is the main reason for the higher investment cost.

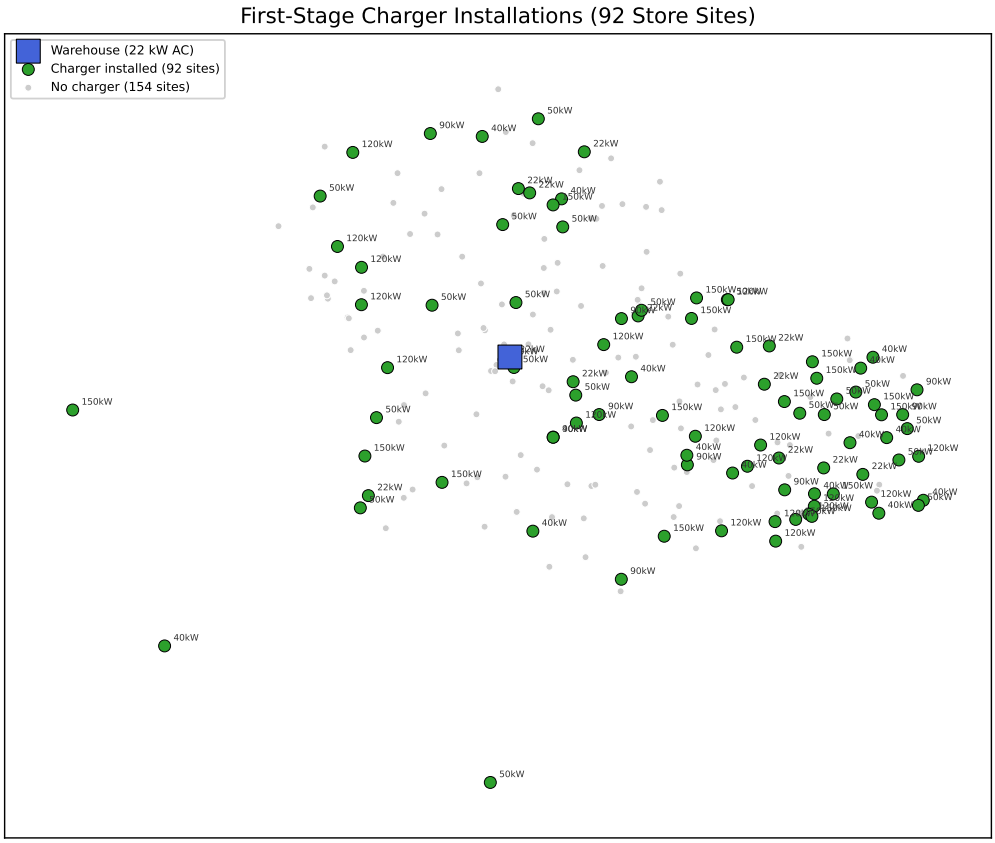

Figure 8.6: Installed store chargers in the main experiment. Labels show charger power.

Evaluation

When we tested this plan against all scenarios, the highest daily cost was about 17,379 EUR, and the average cost was about 11,463 EUR. This is a significant improvement over the results from the smaller clusters. These findings clearly show that using larger clusters leads to a much better and more efficient plan.

Final corrections

In this experiment, reoptimization was not needed. The plan found in Stage 1 was already strong enough that no further improvements were possible. This is because the larger clusters were already able to identify and handle the most difficult scenarios during the initial calculation. This confirms that the larger cluster size avoids the weaknesses seen in the smaller groups.

8.4.4 Cluster-Size Comparison

The table below compares the results from both experiments. It is important to distinguish between the one-time cost of buying and installing the chargers and the daily cost of making deliveries. Our final results focus on reducing this daily delivery cost.

Table 8.3: Comparison of the completed full-instance runs. Runtime is wall-clock time per stage. Costs are EUR per worst evaluated scenario.

Metric	12 stores per cluster	Main 24-stores-per-cluster run
Stores	259	259
Scenarios	31	31
Clusters	22	11
Installed store chargers	94	95
Store-charger investment cost	3,622,000	4,805,000
Stage 1 wall runtime (h)	5.22	54.51
Stage 2 wall runtime (h)	2.24	1.81
Stage 3 wall runtime (h)	27.77	0.11
Stage 2 worst-case cost	23,278.78	17,378.72
Final worst-case cost	22,100.73	17,378.72
Stage 3 improvement	5.06 %	0.00 %

Investment cost decomposition. Both runs install almost the same number of chargers (94 vs. 95), yet the 24-stores run costs roughly EUR 1.2 million more in capex. Figure 8.7 explains the gap: the 12-stores run leans heavily on cheap 22 kW AC chargers (37 of 94, about 40 %), while the 24-stores run shifts the bulk of installations to medium- and high-power DC chargers (56 of 95 chargers at 50 kW or above, vs. only 29 in the 12-stores run). This is consistent with the cost-sensitivity finding (Section 8.2): once routes get longer and waiting time accumulates, fast DC chargers earn back their higher unit price by reducing driver-waiting cost. Larger clusters therefore not only need to be solved differently but also call for a structurally different charging infrastructure.

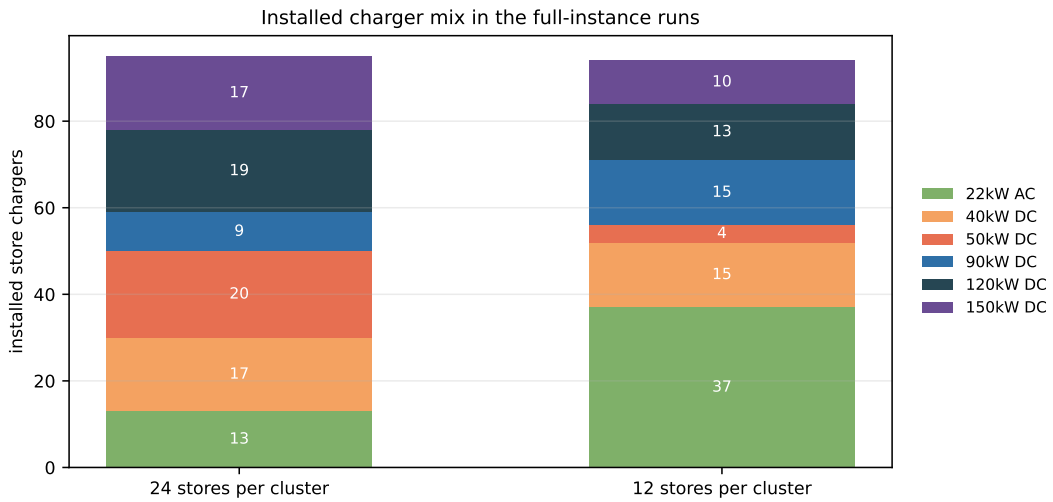

Figure 8.7: Installed charger mix in both full-instance runs. Total counts are nearly identical (94 vs. 95), but the type distribution differs sharply.

Site-level overlap. Comparing the two plans store by store reveals that the structural difference goes beyond charger types. Of the 92 sites equipped in the 24-stores run and the 91 sites in the 12-stores run, only 42 are shared (about 46 %); a further 50 sites are unique to the 24-stores plan and 49 are unique to the 12-stores plan. Even among the 42 shared locations, only 5 receive the same charger type in both runs. The cluster boundaries thus determine not just *how many* chargers are placed but also *which stores* are chosen, confirming that cluster size is a first-order design decision rather than a tunable performance parameter.

These results show a clear trade-off. Using small clusters is faster at first but leads to a much higher final cost and a less efficient charger plan. In contrast, using larger clusters takes more time to calculate but results in a much better objective value. This is the expected outcome: larger clusters preserve more of the natural delivery structure, allow the solver to share charging capacity across more stores, and offer more routing flexibility to absorb difficult demand scenarios. Our final recommendation is therefore not a fixed cluster size but a guideline: the cluster size should be chosen based on the hardware the algorithm is running on, picking the largest size that still solves within the available time budget. To support this choice in practice, the web application (Section 10.3) provides a runtime estimate based on the chosen number of stores, clustering method, and processor cores, so users can size their clusters before starting a run. The 24-stores-per-cluster run, which achieved a daily delivery cost of 17,379 EUR, is the largest size we could solve on our setup; the smaller experiment is included only to show how cluster size affects the quality of the results.

9 Discussion

9.1 Scope and Limitations

Our model is designed for high-level planning. It helps decide where to install chargers and ensures the plan works well on different days. It is not intended for detailed, daily truck scheduling. Please keep the following limitations in mind:

- *Fixed cluster boundaries.* Trucks cannot travel between clusters, and these groupings cannot be changed once they are set.
- *Independent days.* Each day is treated as a separate event, so we do not consider how demand from one day might affect the next.
- *Time-limited calculations.* Because we used time limits for the calculations, our results are high-quality practical solutions rather than mathematically perfect answers.
- *Simplified warehouse charging.* We assumed a fixed charging setup at the warehouse and did not explore other options like faster chargers or power grid limits.
- *Fixed travel times.* We used fixed distances and travel times, which do not account for traffic, weather, or road construction.
- *No specific time windows.* The model does not consider store opening hours or exact appointment times.
- *No battery aging costs.* We do not include the cost of batteries losing capacity over time as they are used and charged.

Despite these limitations, our results provide valuable insights for infrastructure planning. The goal of this research is to answer high-level questions about where to place chargers rather than solving every detail of daily operations.

9.2 What the Full-Instance Runs Show

Our main experiment using 24 stores per cluster produced the best results. The final plan recommends 95 chargers and results in a daily operating cost of EUR 17 378.72. This is the primary finding of our study.

The experiment with 12 stores per cluster was faster but less effective. While it makes Stage 1 much easier to solve, it results in a higher final daily cost because breaking the map into small pieces removes too many routing options. This clearly shows that cluster size is not just a speed dial; it is a major trade-off between speed and quality.

This smaller experiment also highlighted the importance of our final correction stage (Stage 3). Its initial cost was EUR 23 278.78, but Stage 3 improved this to EUR 22 100.73. In the main experiment, Stage 3 was not needed because the global worst scenario was already handled by almost all clusters during Stage 1. This proves that larger clusters are better at identifying difficult scenarios early on. Stage 3 acts as a final safety check that ensures the plan is reliable.

9.3 The Role of Clustering

Clustering is essential for making the problem solvable, but it also introduces some limitations. By splitting the stores into groups, we can solve the problem more easily, but we also restrict delivery routes. The 96-store test supported using demand-aware angular slices as the best choice. While the geographic method came close in cost, it created a cluster with only one store, which is not ideal for general planning. We also found that having an equal number of stores in each group (the store-count method) actually resulted in the highest total cost.

These findings are specific to this retailer's layout, where stores are centered around a single warehouse. If the network changes, different methods might perform better. We recommend re-evaluating the clustering strategy whenever the map changes.

9.4 Cost Factors

Our tests showed that the cost of driver waiting time and the price of chargers have the biggest impact on our plan. Small changes in electricity prices did not significantly change where we placed the chargers. This is good news for planners, as they can focus on getting accurate costs for labor and equipment without worrying too much about fluctuating energy prices. While this test was limited to a single 16-store cluster, it identifies the most important factors for future planning.

9.5 Practical Benefits of Store Charging

The idea of charging at stores is a practical solution for electric trucks. Since trucks are already stopped at stores to unload cargo, they can charge during this time without losing efficiency. This also reduces the need for large, expensive charging setups at the main warehouse.

Our final plan is a strong recommendation, but real-world factors must still be considered. Before building charging stations, a company would need to check power grid capacity, land ownership, and local regulations. Our model identifies the best locations for these chargers based on their operational value.

9.6 Future Work

Several future improvements could make our results even stronger:

- Compare clustering methods on the full 259-store problem to see if the ranking holds true at large scale.
- Test how cost sensitivity affects the entire network rather than just one cluster.
- Include more details about charging at the warehouse, such as power grid limits, peak-demand charges, and waiting lines.
- Allow Stage 3 to be more flexible, for example by moving stores between clusters or merging nearby clusters temporarily.
- Include more realistic factors like exact delivery time windows, unpredictable traffic delays, and actual battery aging costs.

This thesis provides a complete process for planning charging infrastructure. While there is always more work to be done, we have proven that this method works for large-scale, real-world problems.

10 An Operational Tool for Retailers

While our research focused on one retailer in southern Germany, we also created a software tool that anyone can use. This tool simplifies the complex optimization process into a straightforward workflow. Any company can upload their delivery data and get a customized plan. This chapter introduces the tool and explains how it helps retailers switch to electric trucks more easily.

10.1 Motivation

Planning for chargers is an ongoing process. Companies need to update their plans as they open new stores, as energy prices change, or as better technology becomes available. Performing these complex calculations manually each time is not practical. Our web application makes this easy. Any planner can now create a test case, run the optimization, and review the results without needing to write any code or understand the underlying math.

10.2 Designing a New Experiment

Creating a new test case is a simple three-step process:

1. **Source:** Choose a data file or upload a new one.
2. **Filtering:** Select which stores to include based on their distance from the warehouse.
3. **Clustering:** Divide the stores into groups. The tool provides a live preview on a map so that planners can quickly test different group sizes and methods until they find the best setup.

10.3 Running the Solver and Tracking Progress

Once a case is saved, the user sets the parameters and starts the optimization. The backend starts the run, stores the results, and monitors the background jobs. This can run on a local computer or a more powerful remote server. While the computer is working, the

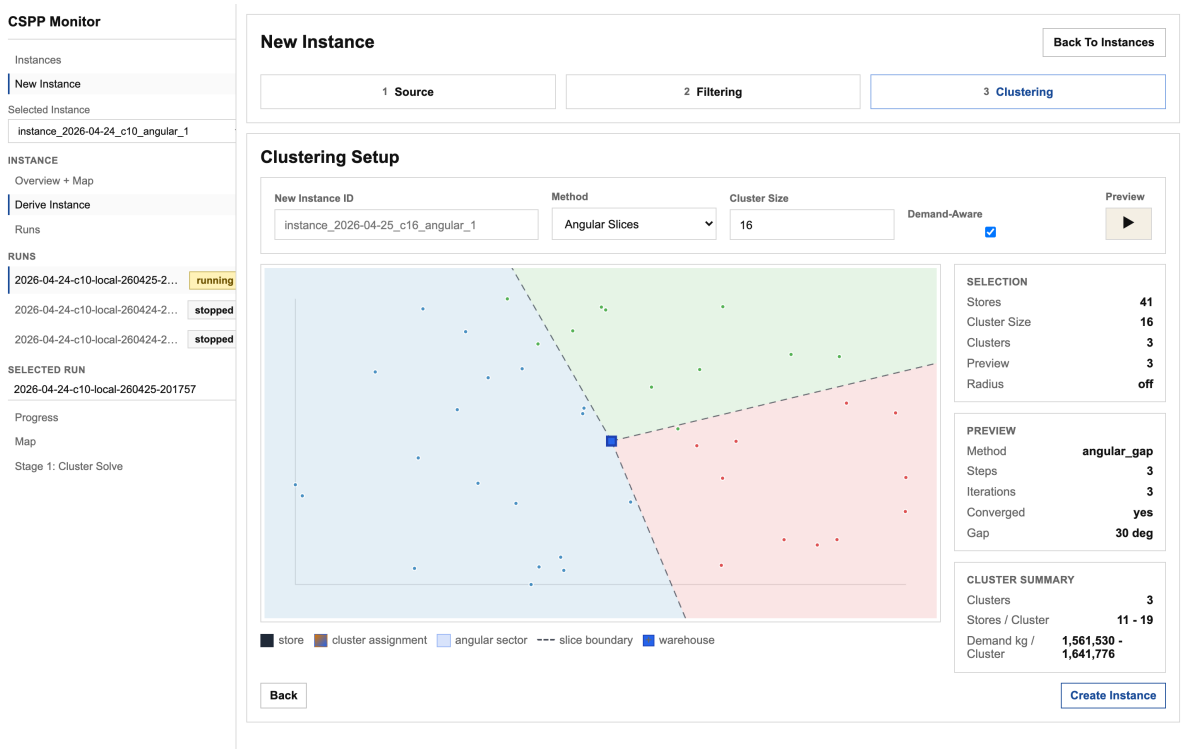

Figure 10.1: Step three of the new-instance builder. Planners can preview different grouping methods on the map and see important data about the result before starting the calculation.

frontend progress panel shows the current stage, the time elapsed, and the estimated total time remaining based on the number of processor cores being used. The application calculates this estimate by checking the number of stores and the clustering method, comparing them to historical data, and adjusting for any extra calculation rounds. The tool also provides real-time charts for each group of stores, showing how the cost improves as the solver works.

10.4 Reviewing the Final Charger Plan

When the calculation is complete, the results are displayed on an interactive map. Users can see where the chargers should be placed and view the planned delivery routes for different days. The tool also highlights any changes made during the final correction stage. For those who want more detail, separate pages provide in-depth charts and data for every part of the plan.

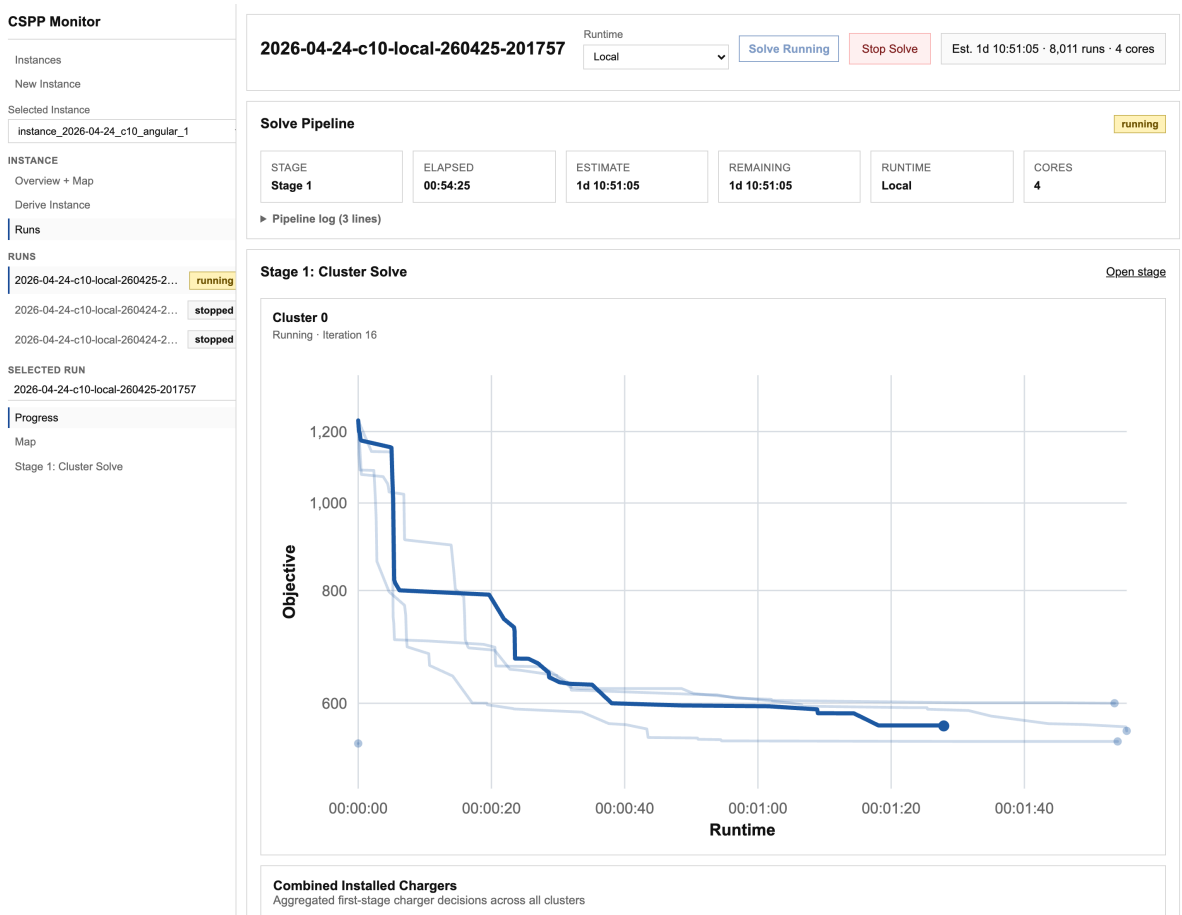

Figure 10.2: Progress view while the optimization is running. The screen shows the estimated time remaining based on processor cores and provides detailed charts for each cluster.

10.5 A Tool for Any Retailer

The most important part of this tool is its flexibility. It can be used by any retailer, regardless of their location or data. Users can easily adjust costs, such as electricity prices and truck expenses, to fit their specific needs. The true value of this research is not just one charger plan, but a powerful tool that allows any company to create a reliable, data-driven plan for their entire delivery network in just a few minutes.

11 Conclusion and Outlook

11.1 Conclusion

This thesis created and tested a reliable process for planning where to place chargers for electric grocery trucks. The process uses real-world data to identify demand patterns, groups stores into clusters, determines the best locations for chargers, and ensures the plan works well even on the most difficult days.

Our main experiment using 24 stores per cluster resulted in a plan with 95 chargers and a daily delivery cost of 17,379 EUR. We found that using larger clusters leads to better results, although they take more time to solve. For this retailer, the 24-store clusters were the best choice, but different sizes might be appropriate for other networks.

Our tests showed that grouping stores by their location and demand works best. We also identified that the cost of driver waiting time and the price of chargers are the most important economic factors. Most importantly, we developed a web application that makes this complex optimization process accessible to any retailer. Companies can now use their own data to quickly calculate the costs and benefits of switching to electric trucks.

11.2 Answers to the Research Questions

1. **How can historical retailer data be transformed into a reliable plan?** We developed a process that turns raw delivery records into store coordinates, demand scenarios, unloading-time guesses, and route data. These are fed into a robust model where charger decisions are made first, while routing decisions adjust to the actual demand of each day.

2. **How can realistic parameter values be obtained?** The technical settings come directly from vehicle and charger spec sheets, while daily operational numbers come from the retailer's records and planning rules. We found that focus should be placed on accurate costs for labor and equipment, as these have the biggest impact on the final plan.

3. **Which choices make the problem practically solvable?** Because the full problem is too massive for a single calculation, the practical solution is to use clustering, time-limited

solving, scenario checking, and active-set corrections. Our experiments show that this process works, but the size of the clusters is a critical factor in the quality of the final results.

11.3 Future Research

Future research should explore different grouping methods on a larger scale and allow for more flexibility in how stores are assigned to groups. We also recommend adding more realistic factors like traffic delays and exact delivery times. Finally, the web application can be improved with more interactive features to help retailers test different scenarios and compare results more easily. This study provides a strong foundation for the future of electric grocery delivery.

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Acronyms

CSPP Charging Station Placement Problem. iv, 1–3, 10, 29, 30, 39

MIP Mixed-Integer Programming. 29, 30, 33, 34, 39, 42, 43, 67

SoC State of Charge. 7, 19, 27

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Declaration of AI Use

I confirm the use of Artificial Intelligence (AI) tools in the preparation of this thesis. Specifically, I used Gemini, Claude, and OpenAI's Codex for the following purposes:

- **Language Support:** I used these tools for style refinement and grammar correction of my own writing. All AI-suggested phrasing was manually reviewed and rewritten to ensure it accurately reflected my findings.
- **Idea Exploration and Clarification:** AI tools were used to understand the underlying physics of batteries and energy consumption in electric vehicles, and other concepts related to the parameter calibration.
- **Programming and Mathematical Support:** When getting started, AI was helpful to understand how constraints for MIPs are formally written and converted into Python code. AI also assisted with general code completion, debugging, and boilerplate generation. All AI-generated code was manually reviewed, tested, and modified to ensure it met the specific requirements of this thesis. The core logic and structure of the algorithms were designed and implemented by me.

In accordance with the guidelines of the Professorship of Optimization and Sustainable Decision Making, I ensured the accuracy and integrity of the work by manually verifying all AI-generated suggestions. All academic contributions—including the robust CSPP formulation, the experimental design, and the analysis of results—are my own work. No AI-generated text was included verbatim in the final document without manual revision. I take full responsibility for the originality and correctness of the submitted work.